Department of Information and Communication Technology

Faculty of Technology

University of Ruhuna

Object Oriented Programming Practicum

ICT2132

Mini Project

Group NO:

14(Criptixz)

Submitted to: Mr. P.H.P. Nuwan Laksiri

Group Members -

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Brief Introduction about the problem/ Group project

Develop a System to Managing academic and administrative operations for Faculty Of Technology for handling User profiles (Admin,Lecture,Student, Technical Officer) ,managing Course Details, maintaining Undergraduate Marks and grades, tracking Undergraduate Attendance, handling Notices ,Timetables and Medicals.

Our group project aimed to address this problem by developing a Faculty Management System using Java for the front end and MySQL for the backend. This system helps manage all academic activities in one centralized system for the Faculty of Technology at the University of Ruhuna.

Brief introduction to the Solution

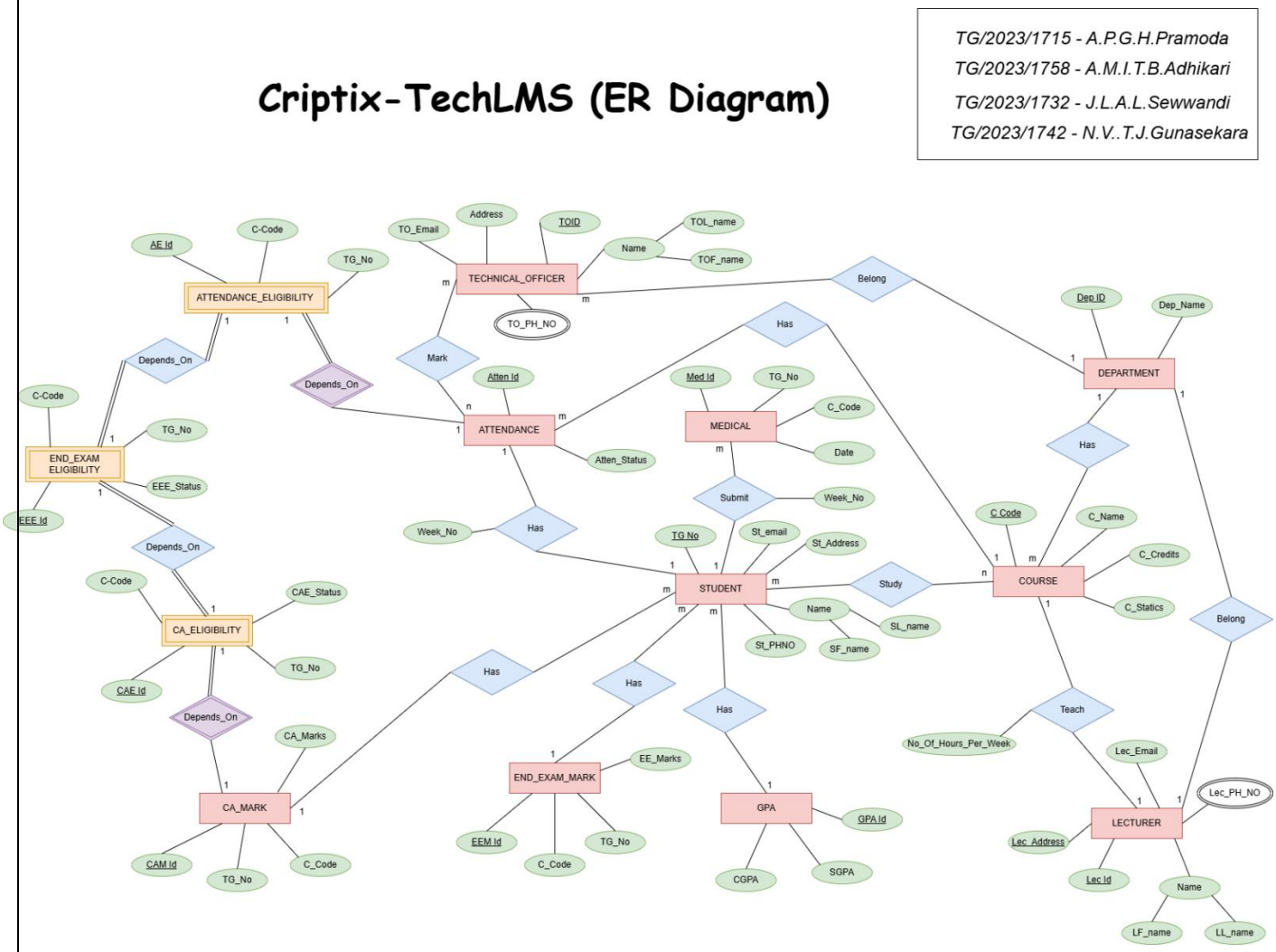
The solution we developed is a Java-based application connected to a MySQL database, designed to support four main types of users:

- ✓ Admin
- ✓ Lecturer
- ✓ Technical Officer
- ✓ Undergraduate

The system automatically identifies the user's role by checking the **prefix of the username** — for example, usernames starting with "**lec**" indicate a Lecturer, "**tg**" for an Undergraduate, "**to**" for a Technical Officer, and "**admin**" for the Admin user. Based on this identification, the system grants access to specific features and interfaces related to their role.

By using a **centralized design**, all operations are streamlined under one system, making academic processes faster, more accurate, and easier to monitor.

ER Diagram of TechLMS



Class Diagram of TechLMS

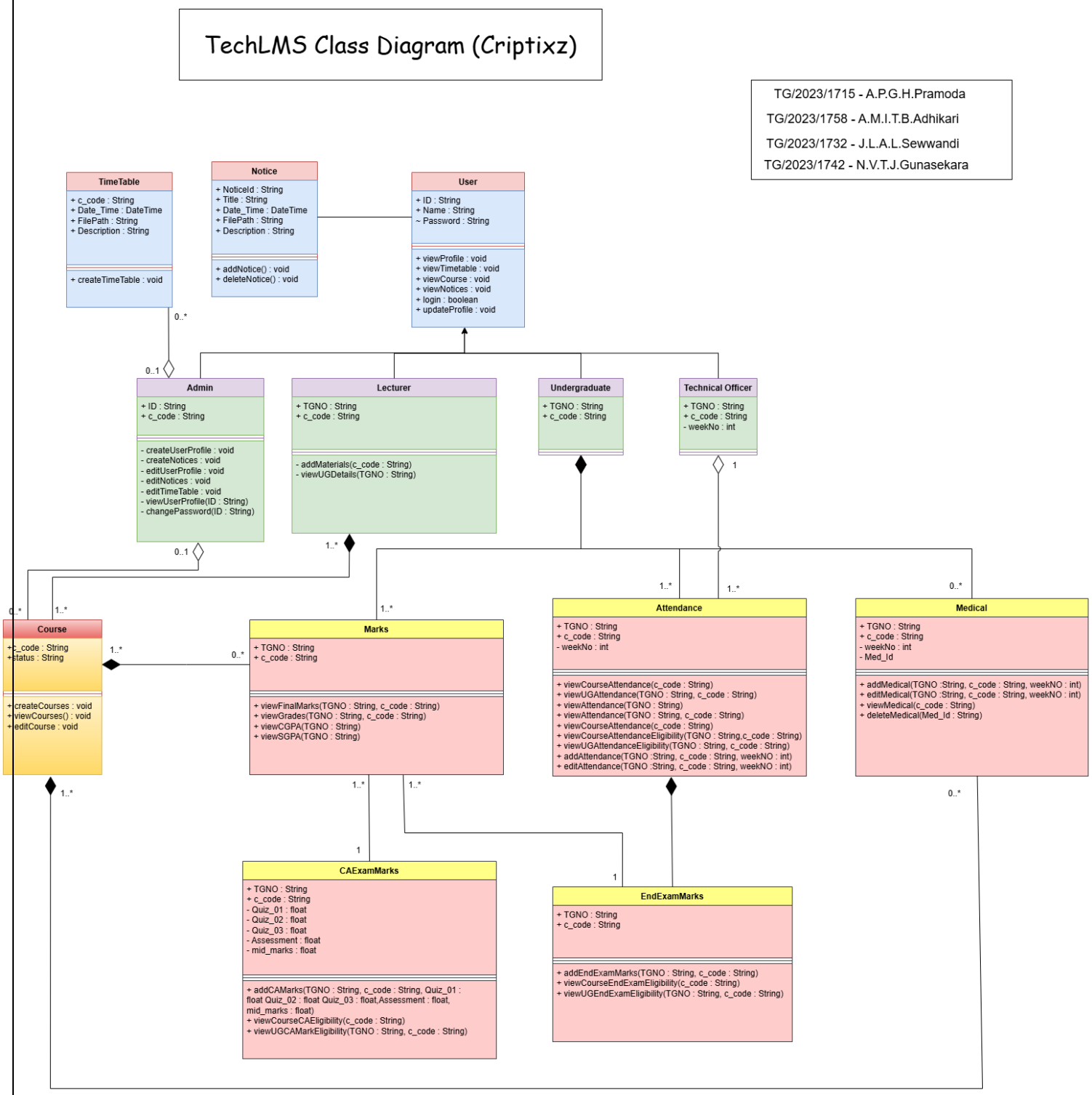


Table solutions in database use in TechLMS

The system uses a MySQL database to store and manage all information, including user profiles, course details, attendance records, medical submissions, marks, grades, timetables, and notices. We designed the database with connected tables using primary and foreign keys to keep the data accurate, consistent, and easy to manage.

Users details

user_id	username	password	role	full_name	email	phone	profile_pic	department	created_at
1	admin	admin123	ADMIN	System Admin	admin@tech.lk	0711000000	NULL	Administration	2026-04-24 23:42:44
2	lec_silva	lec123	LECTURER	Dr. K. Silva	silva@tech.lk	0712000001	NULL	Information & Communication Technology	2026-04-24 23:42:44
3	lec_perera	lec123	LECTURER	Dr. S. Perera	perera@tech.lk	0712000002	NULL	Information & Communication Technology	2026-04-24 23:42:44
4	lec_jayantha	lec123	LECTURER	Mr. A. Jayantha	jayantha@tech.lk	0712000003	NULL	Bio System Technology	2026-04-24 23:42:44
5	lec_kumari	lec123	LECTURER	Ms. R. Kumari	kumari@tech.lk	0712000004	NULL	Engineering Technology	2026-04-24 23:42:44
6	lec_bandara	lec123	LECTURER	Dr. M. Bandara	bandara@tech.lk	0712000005	NULL	Multidisciplinary Studies	2026-04-24 23:42:44
7	to_nimal	to123	TECH_OFFICER	Mr. Nimal	nimal@tech.lk	0713000001	NULL	Information & Communication Technology	2026-04-24 23:42:44
8	to_kumara	to123	TECH_OFFICER	Mr. Kumara	kumara@tech.lk	0713000002	NULL	Bio System Technology	2026-04-24 23:42:44
9	to_saman	to123	TECH_OFFICER	Mr. Saman	saman@tech.lk	0713000003	NULL	Engineering Technology	2026-04-24 23:42:44
10	to_dilani	to123	TECH_OFFICER	Ms. Dilani	dilani@tech.lk	0713000004	NULL	Multidisciplinary Studies	2026-04-24 23:42:44
11	TG231001	stu123	UNDERGRADUATE	Ashan Fernando	ashan@student.lk	0771001001	NULL	Information & Communication Technology	2026-04-24 23:42:44
12	TG231002	stu123	UNDERGRADUATE	Bimal Perera	bimal@student.lk	0771001002	NULL	Information & Communication Technology	2026-04-24 23:42:44
13	TG231003	stu123	UNDERGRADUATE	Chamari Silva	chamari@student.lk	0771001003	NULL	Information & Communication Technology	2026-04-24 23:42:44
14	TG231004	stu123	UNDERGRADUATE	Dinesh Kumar	dinesh@student.lk	0771001004	NULL	Information & Communication Technology	2026-04-24 23:42:44
15	TG231005	stu123	UNDERGRADUATE	Eranga Weerasinghe	eranga@student.lk	0771001005	NULL	Information & Communication Technology	2026-04-24 23:42:44
16	TG221006	stu123	UNDERGRADUATE	Fathima Rizvi	fathima@student.lk	0771001006	NULL	Information & Communication Technology	2026-04-24 23:42:44
17	TG221007	stu123	UNDERGRADUATE	Gayan Mendis	gayan@student.lk	0771001007	NULL	Information & Communication Technology	2026-04-24 23:42:44
18	TG232001	stu123	UNDERGRADUATE	Hiruni Jayasekara	hiruni@student.lk	0771002001	NULL	Bio System Technology	2026-04-24 23:42:44
19	TG232002	stu123	UNDERGRADUATE	Isuru Pathirana	isuru@student.lk	0771002002	NULL	Bio System Technology	2026-04-24 23:42:44
20	TG232003	stu123	UNDERGRADUATE	Janaki Dissanayake	janaki@student.lk	0771002003	NULL	Bio System Technology	2026-04-24 23:42:44
21	TG233001	stu123	UNDERGRADUATE	Kasun Rajapaksa	kasun@student.lk	0771003001	NULL	Engineering Technology	2026-04-24 23:42:44
22	TG233002	stu123	UNDERGRADUATE	Lakmali Senanayake	lakmali@student.lk	0771003002	NULL	Engineering Technology	2026-04-24 23:42:44
23	TG233003	stu123	UNDERGRADUATE	Malith Gunasekara	malith@student.lk	0771003003	NULL	Engineering Technology	2026-04-24 23:42:44
24	TG233004	stu123	UNDERGRADUATE	Nadeesha Wijeratne	nadeesha@student.lk	0771003004	NULL	Engineering Technology	2026-04-24 23:42:44
25	TG234001	stu123	UNDERGRADUATE	Oshan Liyanage	oshan@student.lk	0771004001	NULL	Multidisciplinary Studies	2026-04-24 23:42:44
26	TG234002	stu123	UNDERGRADUATE	Priyanka Jayawardena	priyanka@student.lk	0771004002	NULL	Multidisciplinary Studies	2026-04-24 23:42:44
27	TG234003	stu123	UNDERGRADUATE	Qasim Mohamed	qasim@student.lk	0771004003	NULL		

```
|      28 | TG211008     | stu123   | UNDERGRADUATE | Rashmi Gunawardena   | rashmi@student.lk   | 0771001008 | NULL
    | Information & Communication Technology | 2026-04-24 23:42:44 |
|      29 | TG212004     | stu123   | UNDERGRADUATE | Sachini Ranaweera    | sachini@student.lk  | 0771002004 | NULL
    | Bio System Technology                  | 2026-04-24 23:42:44 |
|      30 | TG213005     | stu123   | UNDERGRADUATE | Thilina Madushanka   | thilina@student.lk  | 0771003005 | NULL
    | Engineering Technology                 | 2026-04-24 23:42:44 |
+---------+--------------+----------+---------------+----------------------+---------------------+------------+---------
```

Marks details

mark_id	ug_id	course_id	exam_type	marks_value	uploaded_by	uploaded_at
1	1	1	CA1	94.70	2	2026-04-24 23:42:44
2	2	1	CA1	91.60	2	2026-04-24 23:42:44
3	3	1	CA1	79.10	2	2026-04-24 23:42:44
4	4	1	CA1	65.70	2	2026-04-24 23:42:44
5	5	1	CA1	76.00	2	2026-04-24 23:42:44
6	6	1	CA1	87.80	2	2026-04-24 23:42:44
7	7	1	CA1	76.30	2	2026-04-24 23:42:44
8	8	1	CA1	63.00	2	2026-04-24 23:42:44
9	9	1	CA1	71.20	2	2026-04-24 23:42:44
10	10	1	CA1	72.00	2	2026-04-24 23:42:44
11	11	1	CA1	91.10	2	2026-04-24 23:42:44
12	12	1	CA1	64.80	2	2026-04-24 23:42:44
13	13	1	CA1	75.80	2	2026-04-24 23:42:44
14	14	1	CA1	89.50	2	2026-04-24 23:42:44
15	15	1	CA1	85.20	2	2026-04-24 23:42:44
16	16	1	CA1	62.50	2	2026-04-24 23:42:44
17	17	1	CA1	81.60	2	2026-04-24 23:42:44
18	18	1	CA1	85.80	2	2026-04-24 23:42:44
19	19	1	CA1	89.30	2	2026-04-24 23:42:44
20	20	1	CA1	94.00	2	2026-04-24 23:42:44
32	1	1	CA2	63.40	2	2026-04-24 23:42:44
33	2	1	CA2	75.60	2	2026-04-24 23:42:44
34	3	1	CA2	92.90	2	2026-04-24 23:42:44
35	4	1	CA2	52.50	2	2026-04-24 23:42:44
36	5	1	CA2	68.80	2	2026-04-24 23:42:44
37	6	1	CA2	91.60	2	2026-04-24 23:42:44

Timetables details

tt_id	course_id	day_of_week	start_time	end_time	location	session_type
1	1	MON	08:00:00	10:00:00	Hall A - Block 1	THEORY
2	1	MON	10:00:00	12:00:00	Lab 01 - Block 2	PRACTICAL
3	2	TUE	08:00:00	10:00:00	Hall B - Block 1	THEORY
4	2	TUE	13:00:00	15:00:00	Lab 02 - Block 2	PRACTICAL
5	3	WED	08:00:00	10:00:00	Hall C - Block 1	THEORY
6	4	WED	10:00:00	12:00:00	Hall D - Block 3	THEORY
7	4	WED	13:00:00	15:00:00	Lab 03 - Block 3	PRACTICAL
8	5	THU	08:00:00	10:00:00	Hall E - Block 2	THEORY
9	5	THU	13:00:00	15:00:00	Lab 04 - Block 2	PRACTICAL
10	6	MON	13:00:00	15:00:00	Hall F - Block 4	THEORY
11	7	TUE	10:00:00	12:00:00	Hall G - Block 4	THEORY
12	7	TUE	15:00:00	17:00:00	Lab 05 - Block 4	PRACTICAL
13	8	FRI	08:00:00	10:00:00	Hall H - Block 5	THEORY
14	9	FRI	10:00:00	12:00:00	Hall H - Block 5	THEORY

Medicals details

medical_id	ug_id	from_date	to_date	reason	doc_path	is_approved	status	submitted_at
1	1	2026-01-15	2026-01-17	[MED/2026/001] Fever and flu symptoms	MED_TG231001_001.pdf	1	PENDING	2026-04-24 23:42:44
2	1	2026-02-10	2026-02-10	[MED/2026/002] Dental appointment	MED_TG231001_002.pdf	1	PENDING	2026-04-24 23:42:44
3	2	2026-03-05	2026-03-06	[MED/2026/003] Eye infection treatment	MED_TG231002_001.pdf	0	PENDING	2026-04-24 23:42:44
4	3	2026-02-18	2026-02-19	[MED/2026/004] Stomach pain	MED_TG231003_001.pdf	0	PENDING	2026-04-24 23:42:44
5	8	2026-02-20	2026-02-21	[MED/2026/005] Food poisoning	MED_TG232001_001.pdf	1	PENDING	2026-04-24 23:42:44
6	11	2026-03-10	2026-03-11	[MED/2026/006] High fever	MED_TG233001_001.pdf	0	PENDING	2026-04-24 23:42:44
7	18	2026-01-20	2026-01-22	[MED/2026/007] Chronic back pain treatment	MED_TG211008_001.pdf	1	PENDING	2026-04-24 23:42:44

Attendance details

Additionally, many to many relationship tables are included in database. If users update their details then database is automatically updated.

att_id	ug_id	course_id	session_date	session_type	is_present
1	1	1	2026-01-05	THEORY	1
2	2	1	2026-01-05	THEORY	1
3	3	1	2026-01-05	THEORY	1
4	4	1	2026-01-05	THEORY	1
5	5	1	2026-01-05	THEORY	1
6	6	1	2026-01-05	THEORY	1
7	7	1	2026-01-05	THEORY	1
8	8	1	2026-01-05	THEORY	1
9	9	1	2026-01-05	THEORY	1
10	10	1	2026-01-05	THEORY	1
11	11	1	2026-01-05	THEORY	1
12	12	1	2026-01-05	THEORY	1
13	13	1	2026-01-05	THEORY	1
14	14	1	2026-01-05	THEORY	1
15	15	1	2026-01-05	THEORY	1
16	16	1	2026-01-05	THEORY	1
17	17	1	2026-01-05	THEORY	1
18	18	1	2026-01-05	THEORY	1
19	19	1	2026-01-05	THEORY	1
20	20	1	2026-01-05	THEORY	1
32	1	1	2026-01-07	THEORY	1
33	2	1	2026-01-07	THEORY	1
34	3	1	2026-01-07	THEORY	1
35	4	1	2026-01-07	THEORY	1
36	5	1	2026-01-07	THEORY	1
37	6	1	2026-01-07	THEORY	1
38	7	1	2026-01-07	THEORY	1
39	8	1	2026-01-07	THEORY	1
40	9	1	2026-01-07	THEORY	1
41	10	1	2026-01-07	THEORY	1
42	11	1	2026-01-07	THEORY	1
43	12	1	2026-01-07	THEORY	1
44	13	1	2026-01-07	THEORY	1
45	14	1	2026-01-07	THEORY	1
46	15	1	2026-01-07	THEORY	1

Course details

course_id	course_code	course_name	credits	has_theory	has_practical	department	lecturer_id
1	ICT3101	Object Oriented Programming	3	1	1	Information & Communication Technology	2
2	ICT3102	Database Management Systems	3	1	1	Information & Communication Technology	3
3	ICT3103	Computer Networks	3	1	0	Information & Communication Technology	2
4	BIO3101	Biotechnology Fundamentals	3	1	1	Bio System Technology	4
5	BIO3102	Microbiology	3	1	1	Bio System Technology	4
6	ENG3101	Engineering Mathematics	3	1	0	Engineering Technology	5
7	ENG3102	Circuit Analysis	3	1	1	Engineering Technology	5
8	MDS3101	Research Methodology	2	1	0	Multidisciplinary Studies	6
9	MDS3102	Statistics for Technology	2	1	0	Multidisciplinary Studies	6

Notices details

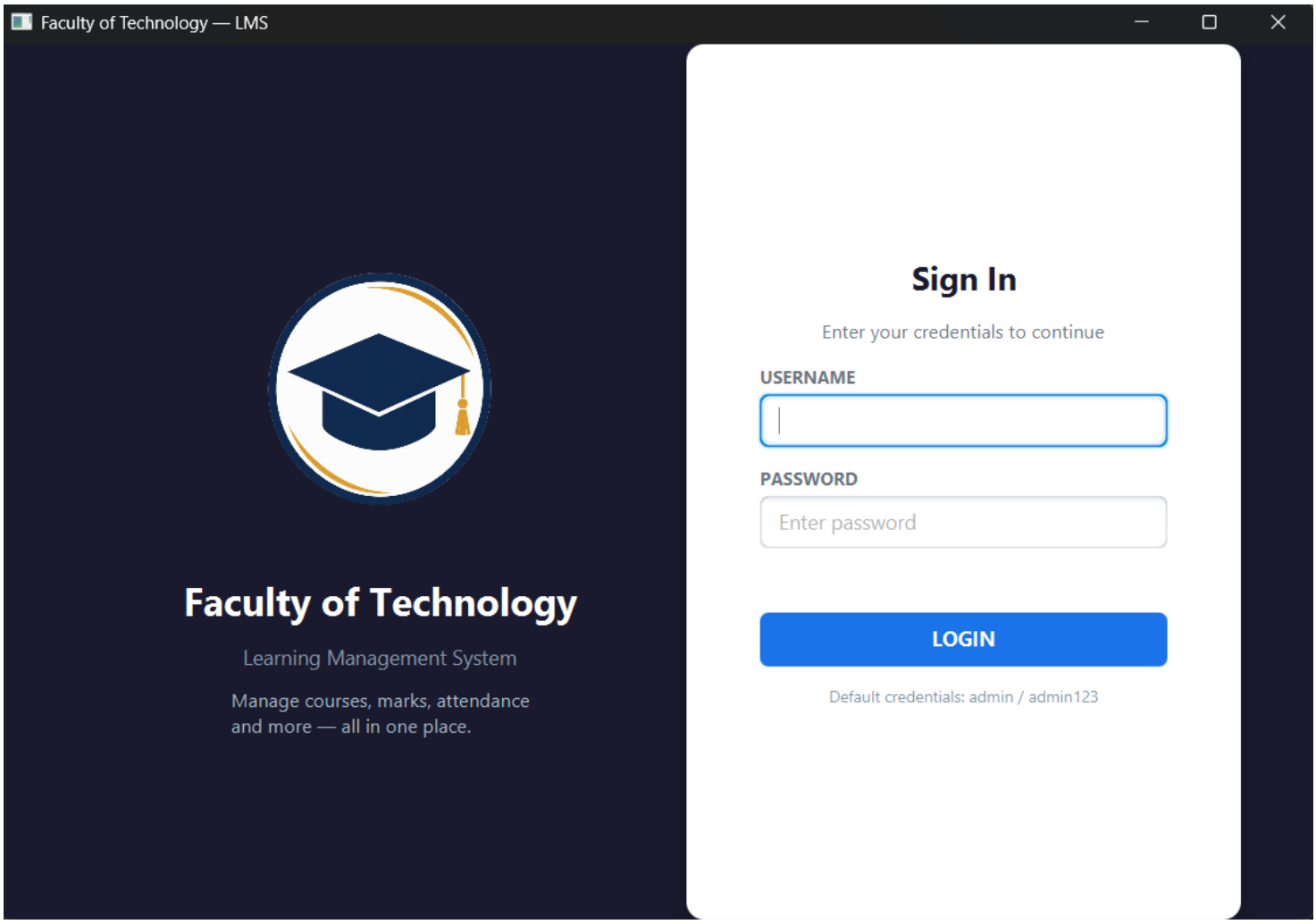
notice_id	title	content	created_by	created_at
1	Semester Exam Schedule Released	The final examination schedule for Semester 1 2026 has been released. Please check the notice board for your exam dates and venues. All students must carry their student ID card.	1	2026-04-24 23:42:44
2	Medical Submission Deadline	All medical certificates for the current semester must be submitted to the Technical Officer before 30th April 2026. Late submissions will not be accepted.	1	2026-04-24 23:42:44
3	Library Closure Notice	The university library will be closed from 15th to 18th April 2026 for annual maintenance. Online resources remain available through the student portal.	1	2026-04-24 23:42:44
4	OOP Lab Sessions Rescheduled	The Object Oriented Programming practical sessions scheduled for 22nd April have been moved to 24th April 2026 in Lab 01. Please update your timetable accordingly.	2	2026-04-24 23:42:44
5	CA Marks Released ? ICT3102	CA1 and CA2 marks for Database Management Systems have been uploaded. Students can view their grades through the student portal.	3	2026-04-24 23:42:44

Undergraduates details

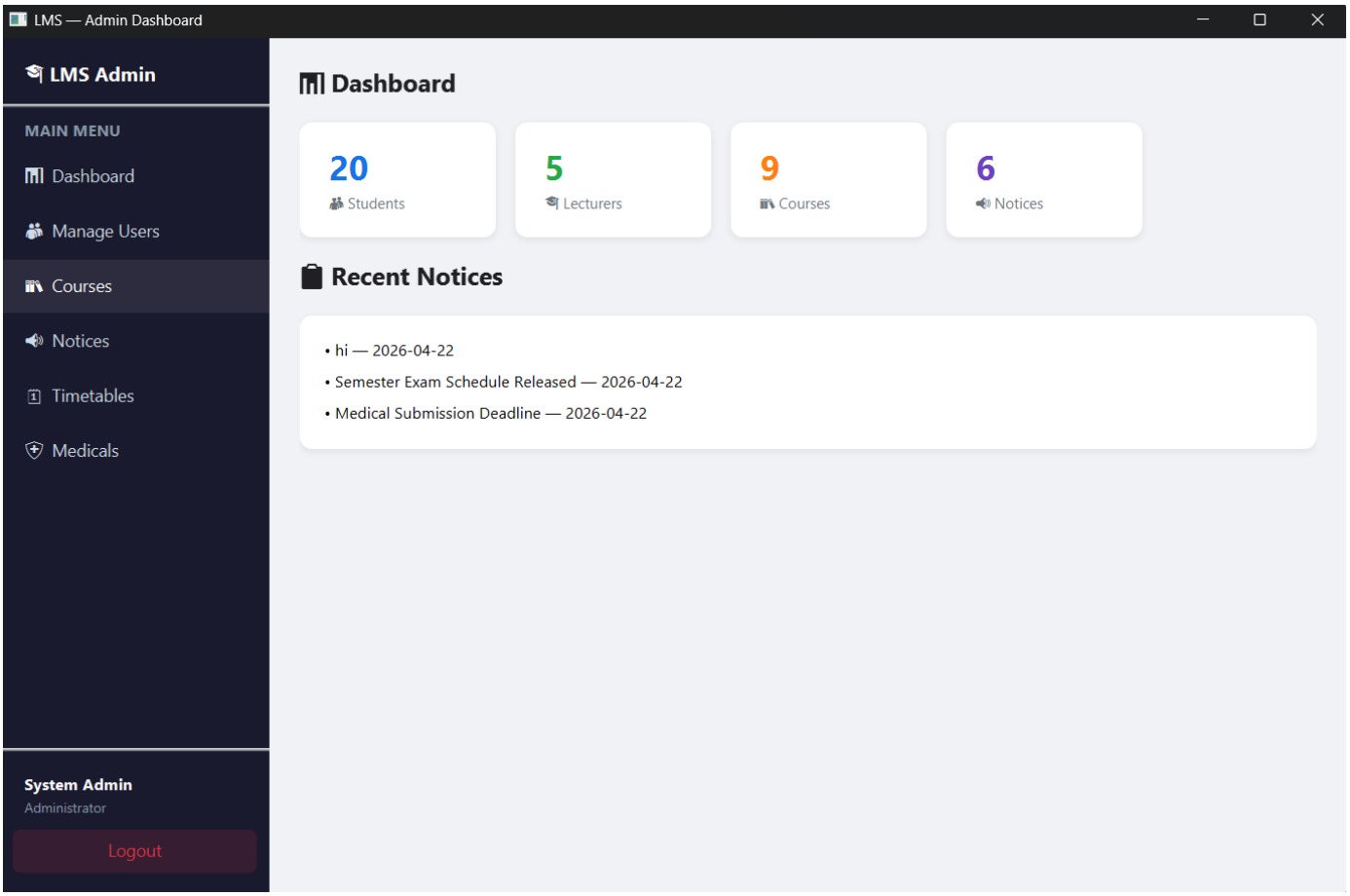
ug_id	user_id	reg_number	batch	is_repeat	batch_missed
1	11	TG/2023/1001	2023	0	0
2	12	TG/2023/1002	2023	0	0
3	13	TG/2023/1003	2023	0	0
4	14	TG/2023/1004	2023	0	0
5	15	TG/2023/1005	2023	0	0
6	16	TG/2022/1006	2022	0	1
7	17	TG/2022/1007	2022	0	1
8	18	TG/2023/2001	2023	0	0
9	19	TG/2023/2002	2023	0	0
10	20	TG/2023/2003	2023	0	0
11	21	TG/2023/3001	2023	0	0
12	22	TG/2023/3002	2023	0	0
13	23	TG/2023/3003	2023	0	0
14	24	TG/2023/3004	2023	0	0
15	25	TG/2023/4001	2023	0	0
16	26	TG/2023/4002	2023	0	0
17	27	TG/2023/4003	2023	0	0
18	28	TG/2021/1008	2021	1	0
19	29	TG/2021/2004	2021	1	0
20	30	TG/2021/3005	2021	1	0

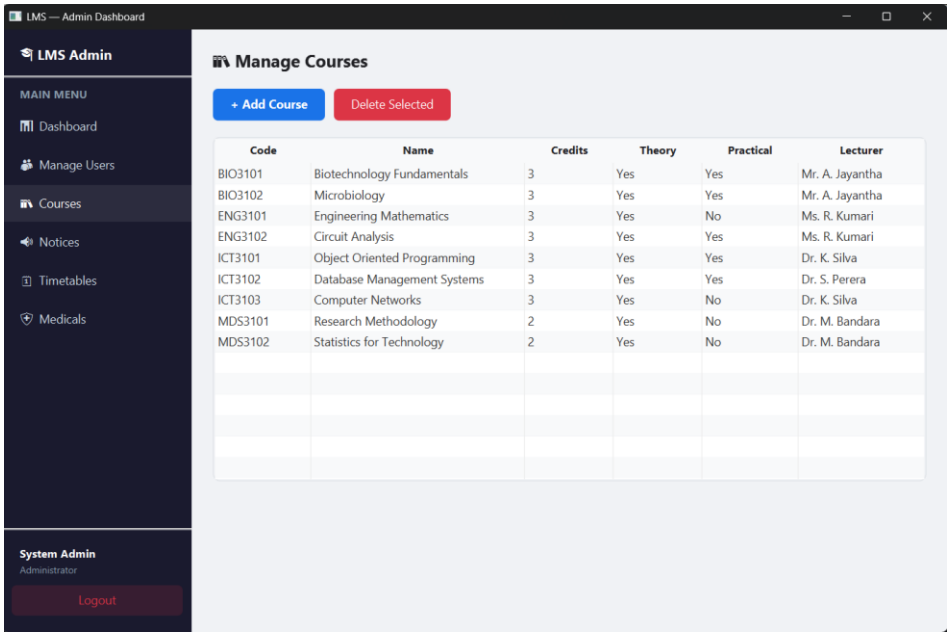
UI designs in TechLMS

Login page

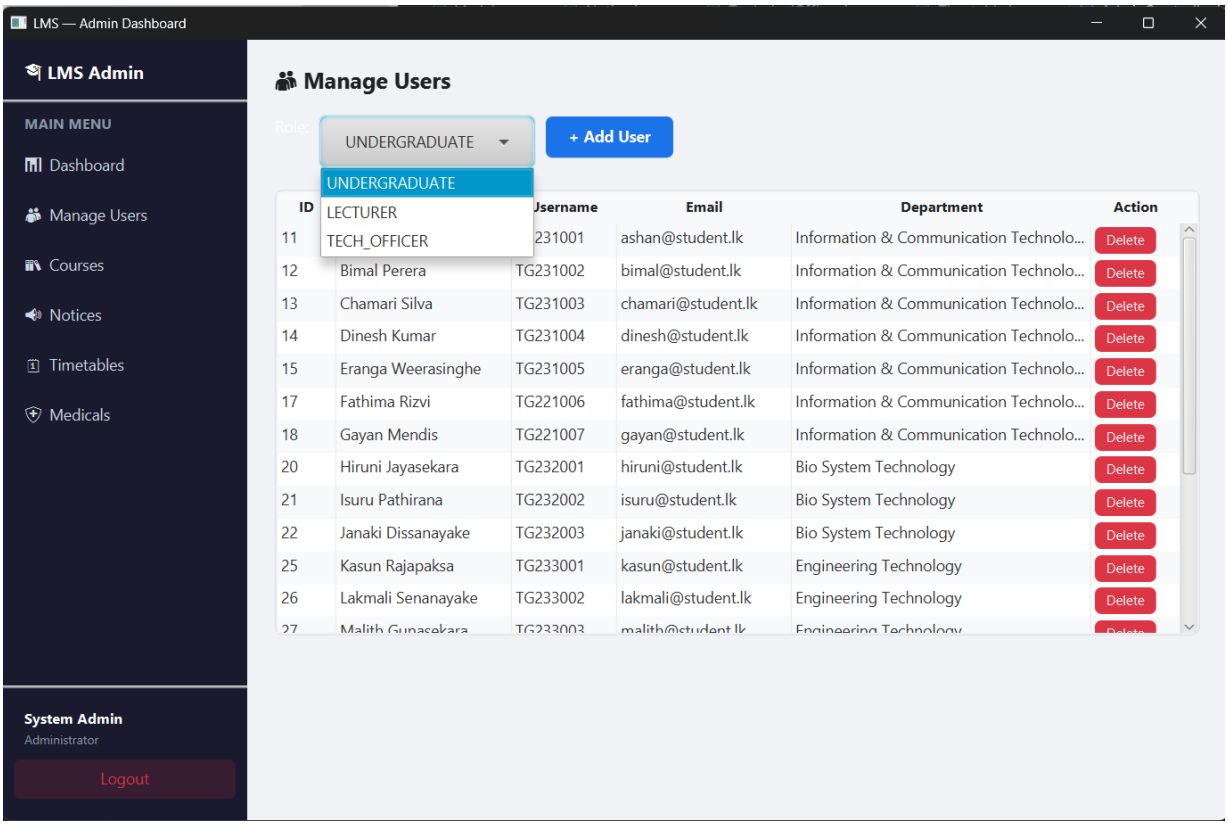


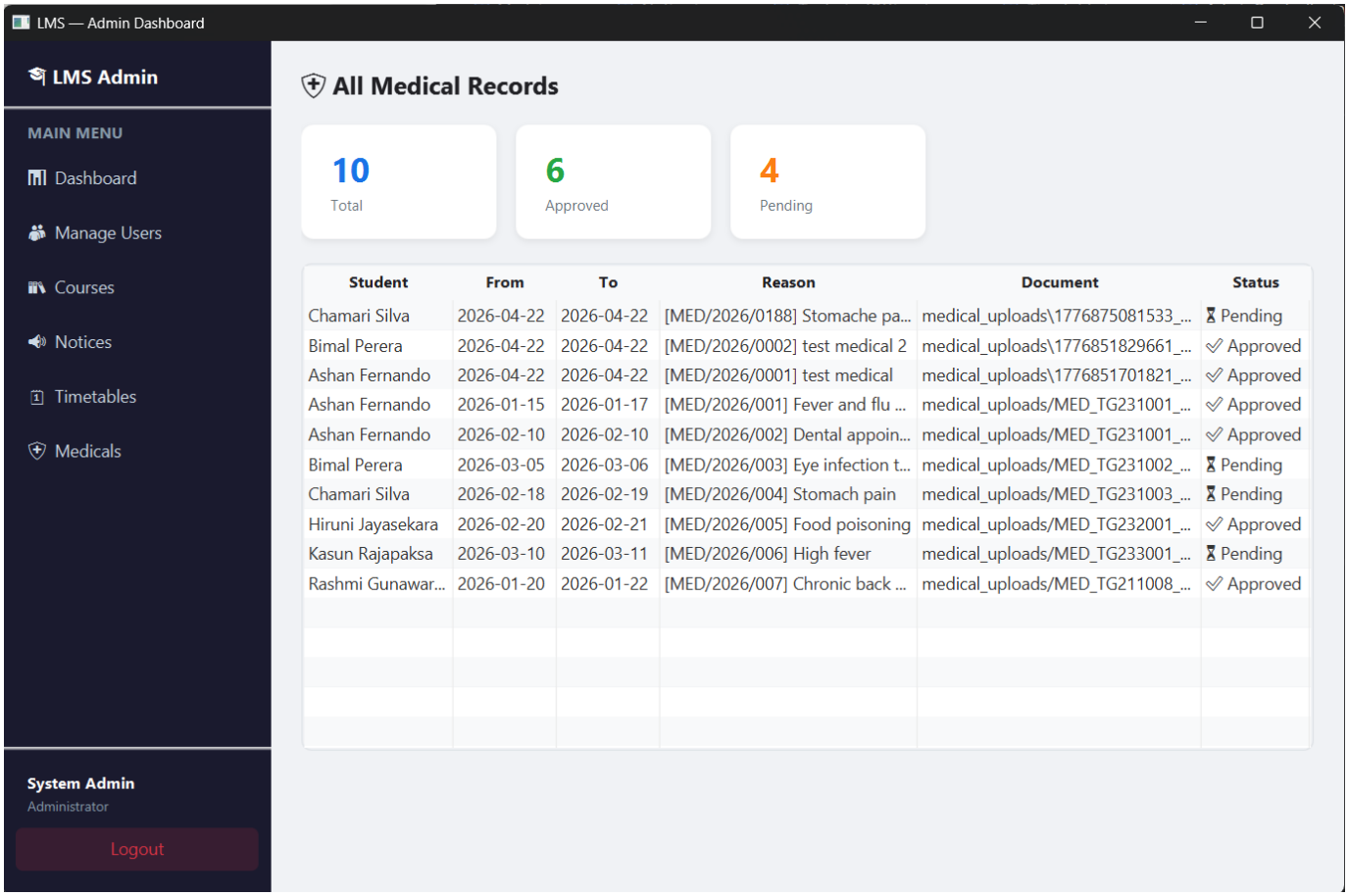
Admin



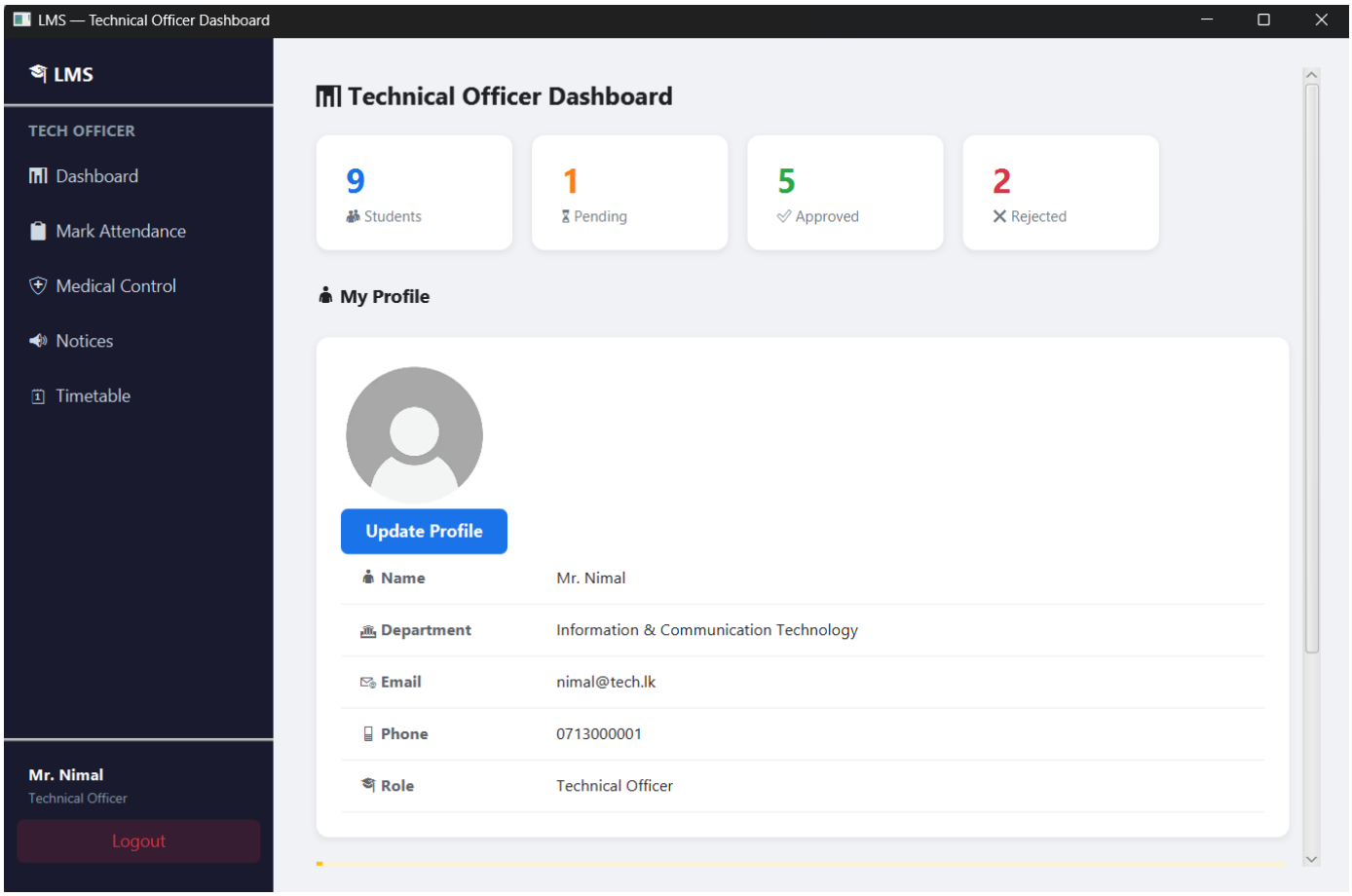


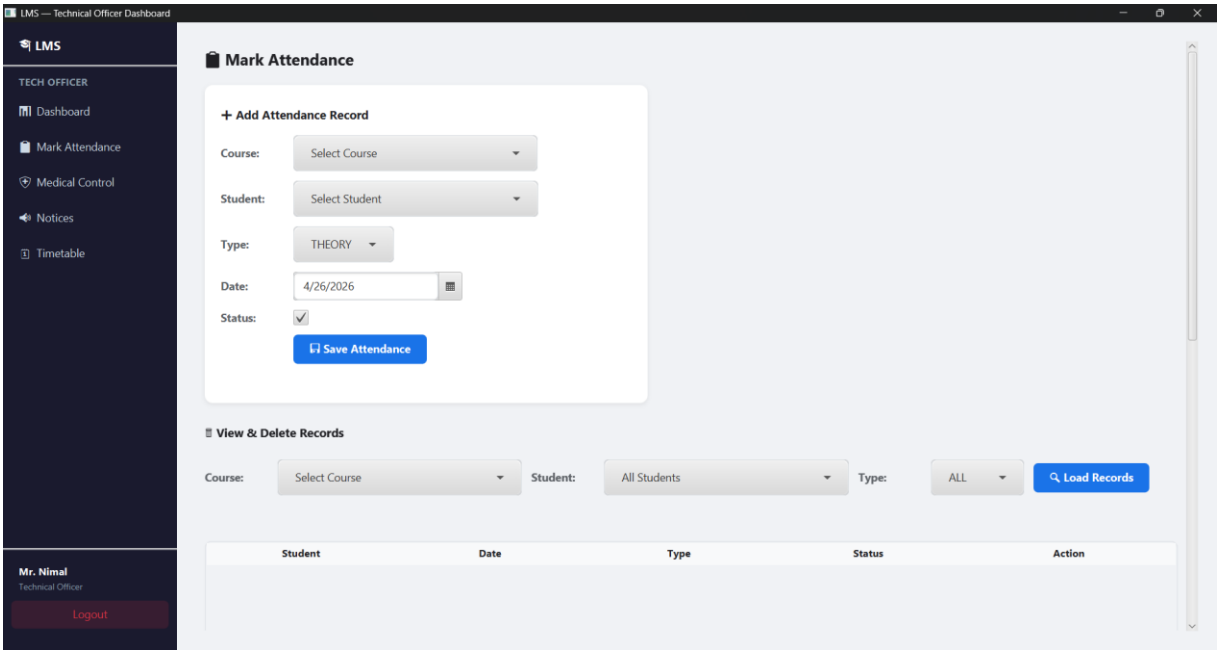
Code	Name	Credits	Theory	Practical	Lecturer
BIO3101	Biotechnology Fundamentals	3	Yes	Yes	Mr. A. Jayantha
BIO3102	Microbiology	3	Yes	Yes	Mr. A. Jayantha
ENG3101	Engineering Mathematics	3	Yes	No	Ms. R. Kumari
ENG3102	Circuit Analysis	3	Yes	Yes	Ms. R. Kumari
ICT3101	Object Oriented Programming	3	Yes	Yes	Dr. K. Silva
ICT3102	Database Management Systems	3	Yes	Yes	Dr. S. Perera
ICT3103	Computer Networks	3	Yes	No	Dr. K. Silva
MDS3101	Research Methodology	2	Yes	No	Dr. M. Bandara
MDS3102	Statistics for Technology	2	Yes	No	Dr. M. Bandara

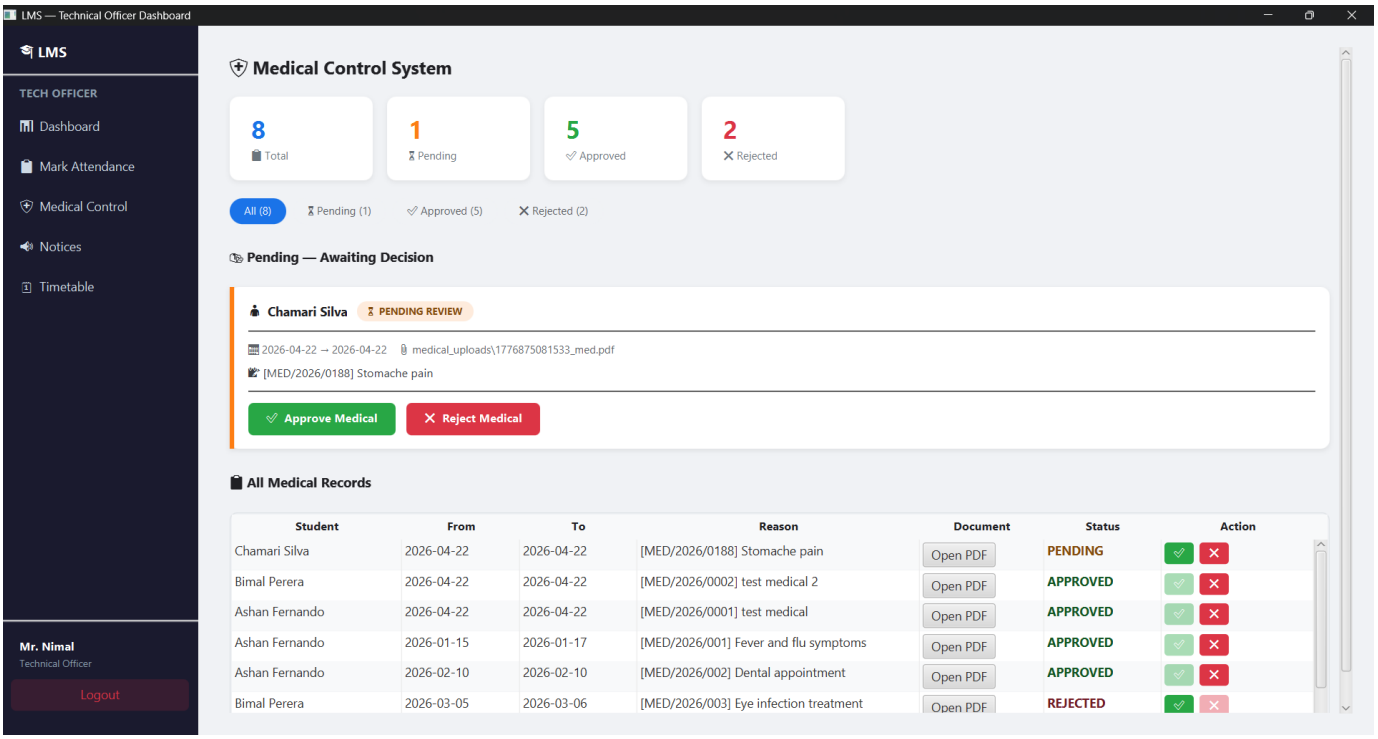

LMS — Admin Dashboard
LMS Admin
MAIN MENU
Dashboard
Manage Users
Courses
Notices
Timetables
Medicals
System Admin
Administrator
Logout
Manage Users
UNDERGRADUATE
+ Add User
UNDERGRADUATE
LECTURER
TECH_OFFICER
ID Username Email Department Action
11 231001 ashan@student.lk Information & Communication Technolo... Delete
12 Bimal Perera TG231002 bimal@student.lk Information & Communication Technolo... Delete
13 Chamari Silva TG231003 chamari@student.lk Information & Communication Technolo... Delete
14 Dinesh Kumar TG231004 dinesh@student.lk Information & Communication Technolo... Delete
15 Eranga Weerasinghe TG231005 eranga@student.lk Information & Communication Technolo... Delete
17 Fathima Rizvi TG221006 fathima@student.lk Information & Communication Technolo... Delete
18 Gayan Mendis TG221007 gayan@student.lk Information & Communication Technolo... Delete
20 Hiruni Jayasekara TG232001 hiruni@student.lk Bio System Technology Delete
21 Isuru Pathirana TG232002 isuru@student.lk Bio System Technology Delete
22 Janaki Dissanayake TG232003 janaki@student.lk Bio System Technology Delete
25 Kasun Rajapaksa TG233001 kasun@student.lk Engineering Technology Delete
26 Lakmali Senanayake TG233002 lakmali@student.lk Engineering Technology Delete


LMS — Admin Dashboard
LMS Admin
MAIN MENU
Dashboard
Manage Users
Courses
Notices
Timetables
Medicals
System Admin
Administrator
Logout
All Medical Records
10
Total
6
Approved
4
Pending
Student From To Reason Document Status
Chamari Silva 2026-04-22 2026-04-22 [MED/2026/0188] Stomache pa... medical_uploads\1776875081533_... Pending
Bimal Perera 2026-04-22 2026-04-22 [MED/2026/0002] test medical 2 medical_uploads\1776851829661_... Approved
Ashan Fernando 2026-04-22 2026-04-22 [MED/2026/0001] test medical medical_uploads\1776851701821_... Approved
Ashan Fernando 2026-01-15 2026-01-17 [MED/2026/001] Fever and flu ... medical_uploads/MED_TG231001_... Approved
Ashan Fernando 2026-02-10 2026-02-10 [MED/2026/002] Dental appoin... medical_uploads/MED_TG231001_... Approved
Bimal Perera 2026-03-05 2026-03-06 [MED/2026/003] Eye infection t... medical_uploads/MED_TG231002_... Pending
Chamari Silva 2026-02-18 2026-02-19 [MED/2026/004] Stomach pain medical_uploads/MED_TG231003_... Pending
Hiruni Jayasekara 2026-02-20 2026-02-21 [MED/2026/005] Food poisoning medical_uploads/MED_TG232001_... Approved
Kasun Rajapaksa 2026-03-10 2026-03-11 [MED/2026/006] High fever medical_uploads/MED_TG233001_... Pending
Rashmi Gunawar... 2026-01-20 2026-01-22 [MED/2026/007] Chronic back ... medical_uploads/MED_TG211008_... Approved

Technical Officer






LMS — Technical Officer Dashboard
LMS
TECH OFFICER
Dashboard
Mark Attendance
Medical Control
Notices
Timetable
Mr. Nimal
Technical Officer
Logout
Medical Control System
8
Total
1
Pending
5
Approved
2
Rejected
All (8)
Pending (1)
Approved (5)
Rejected (2)
Pending — Awaiting Decision
Chamari Silva
PENDING REVIEW
2026-04-22 → 2026-04-22
medical_uploads\1776875081533_med.pdf
[MED/2026/0188] Stomache pain
Approve Medical
Reject Medical
All Medical Records
Student From To Reason Document Status Action
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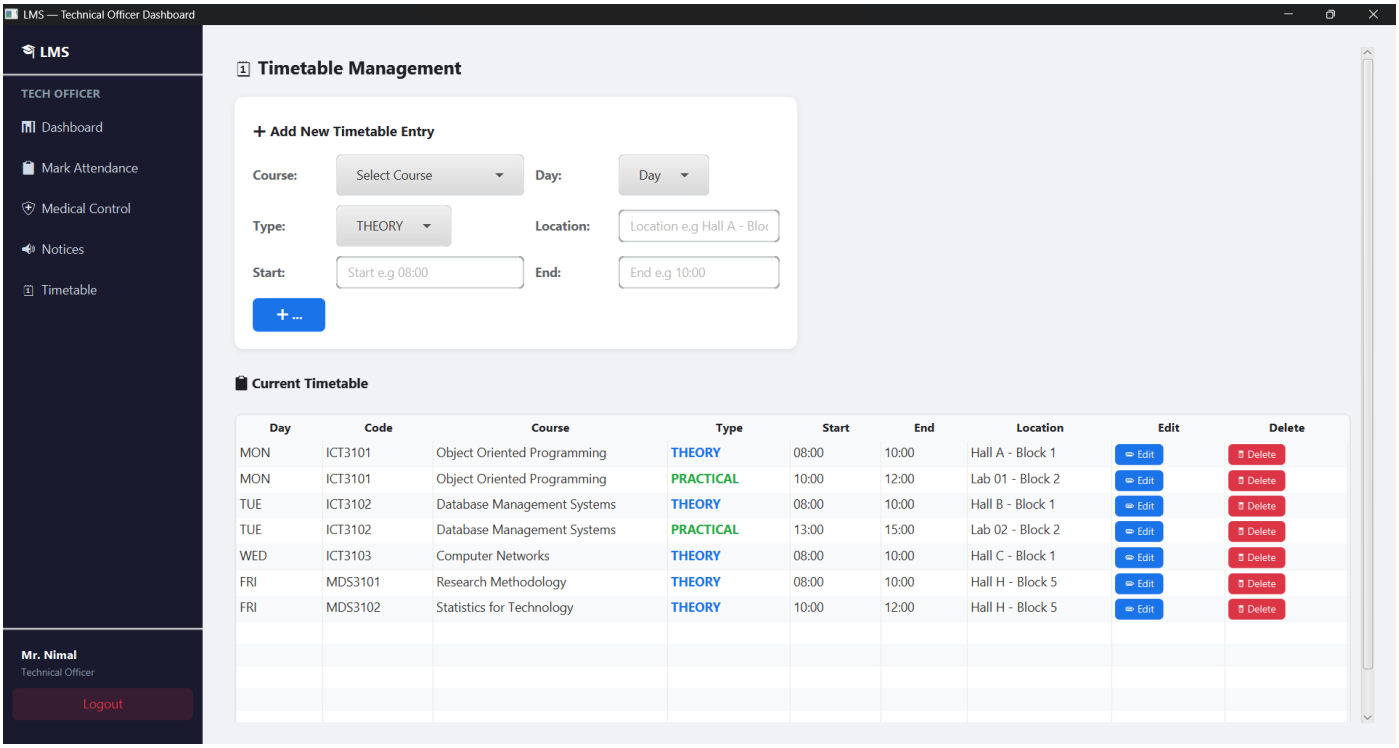
Bimal Perera 2026-04-22 2026-04-22 [MED/2026/0002] test medical 2 Open PDF APPROVED
Ashan Fernando 2026-04-22 2026-04-22 [MED/2026/0001] test medical Open PDF APPROVED
Ashan Fernando 2026-01-15 2026-01-17 [MED/2026/001] Fever and flu symptoms Open PDF APPROVED
Ashan Fernando 2026-02-10 2026-02-10 [MED/2026/002] Dental appointment Open PDF APPROVED
Bimal Perera 2026-03-05 2026-03-06 [MED/2026/003] Eye infection treatment Open PDF REJECTED


LMS — Technical Officer Dashboard
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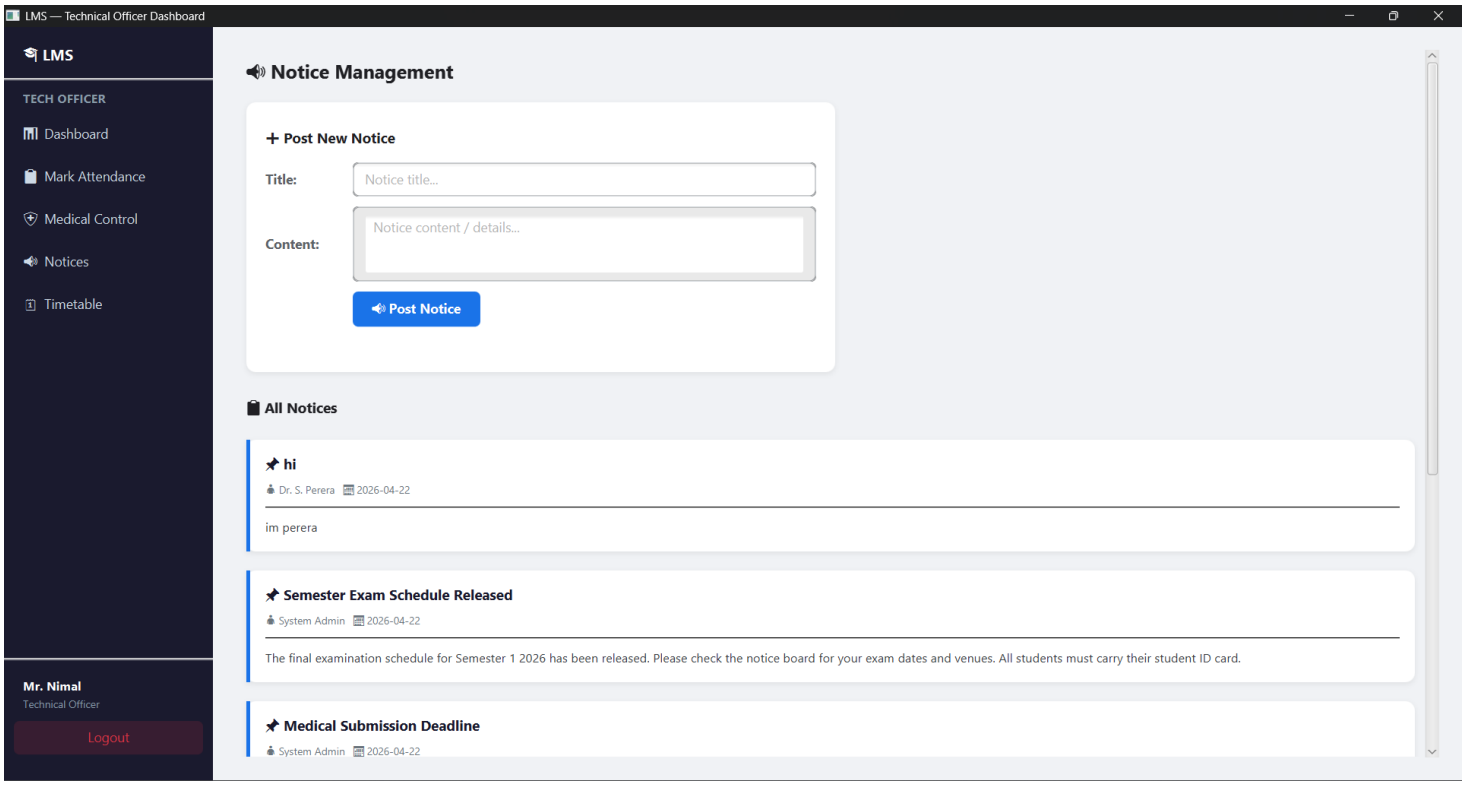
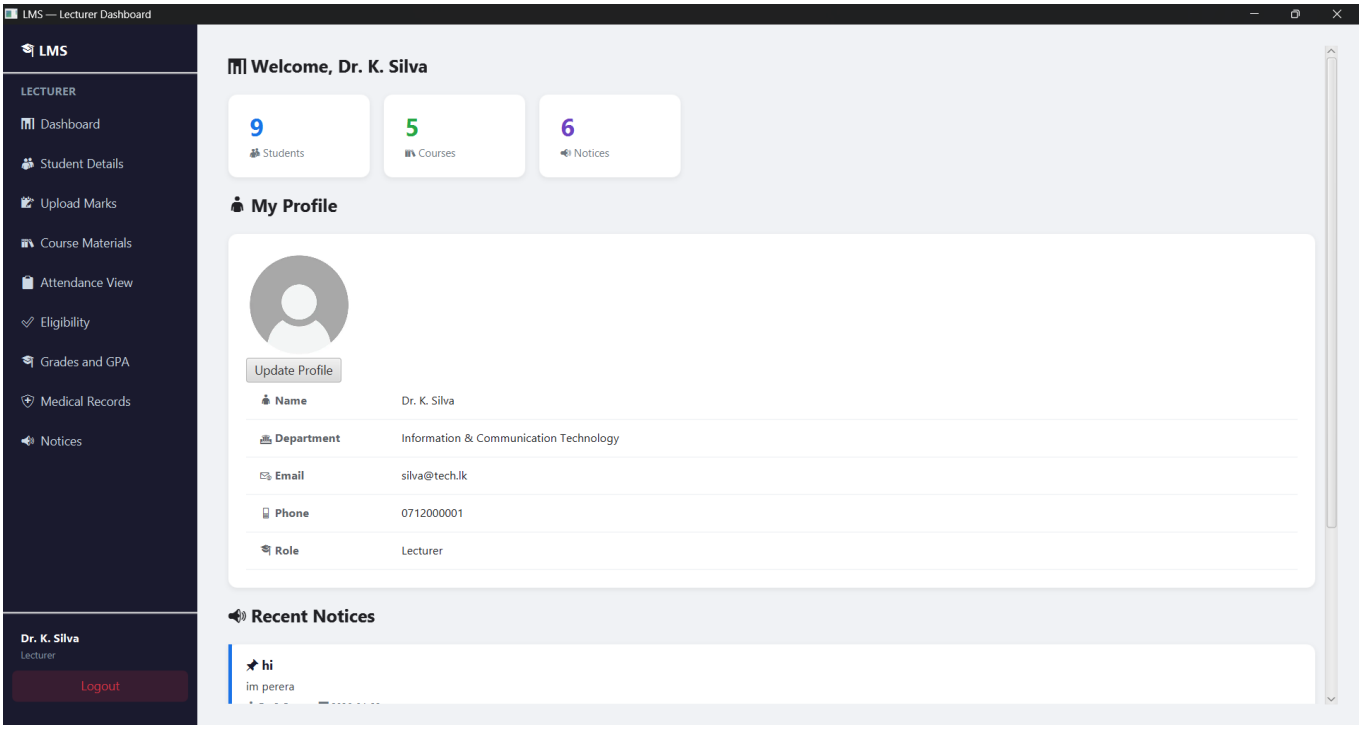
TECH OFFICER
Dashboard
Mark Attendance
Medical Control
Notices
Timetable
Mr. Nimal
Technical Officer
Logout
Timetable Management
+ Add New Timetable Entry
Course: Select Course
Day: Day
Type: THEORY
Location: Location e.g Hall A - Bloc
Start: Start e.g 08:00
End: End e.g 10:00
+ ...
Current Timetable
Day Code Course Type Start End Location Edit Delete
MON ICT3101 Object Oriented Programming THEORY 08:00 10:00 Hall A - Block 1 Edit Delete
MON ICT3101 Object Oriented Programming PRACTICAL 10:00 12:00 Lab 01 - Block 2 Edit Delete
TUE ICT3102 Database Management Systems THEORY 08:00 10:00 Hall B - Block 1 Edit Delete
TUE ICT3102 Database Management Systems PRACTICAL 13:00 15:00 Lab 02 - Block 2 Edit Delete
WED ICT3103 Computer Networks THEORY 08:00 10:00 Hall C - Block 1 Edit Delete
FRI MDS3101 Research Methodology THEORY 08:00 10:00 Hall H - Block 5 Edit Delete
FRI MDS3102 Statistics for Technology THEORY 10:00 12:00 Hall H - Block 5 Edit Delete


LMS — Technical Officer Dashboard
LMS
TECH OFFICER
Dashboard
Mark Attendance
Medical Control
Notices
Timetable
Mr. Nimal
Technical Officer
Logout
Notice Management
+ Post New Notice
Title:
Notice title...
Content:
Notice content / details...
Post Notice
All Notices
hi
Dr. S. Perera 2026-04-22
im perera
Semester Exam Schedule Released
System Admin 2026-04-22
The final examination schedule for Semester 1 2026 has been released. Please check the notice board for your exam dates and venues. All students must carry their student ID card.
Medical Submission Deadline
System Admin 2026-04-22

Lecturer


LMS — Lecturer Dashboard
LMS
LECTURER
Dashboard
Student Details
Upload Marks
Course Materials
Attendance View
Eligibility
Grades and GPA
Medical Records
Notices
Dr. K. Silva
Lecturer
Logout
Welcome, Dr. K. Silva
9
Students
5
Courses
6
Notices
My Profile
Update Profile
Name
Dr. K. Silva
Department
Information & Communication Technology
Email
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Phone
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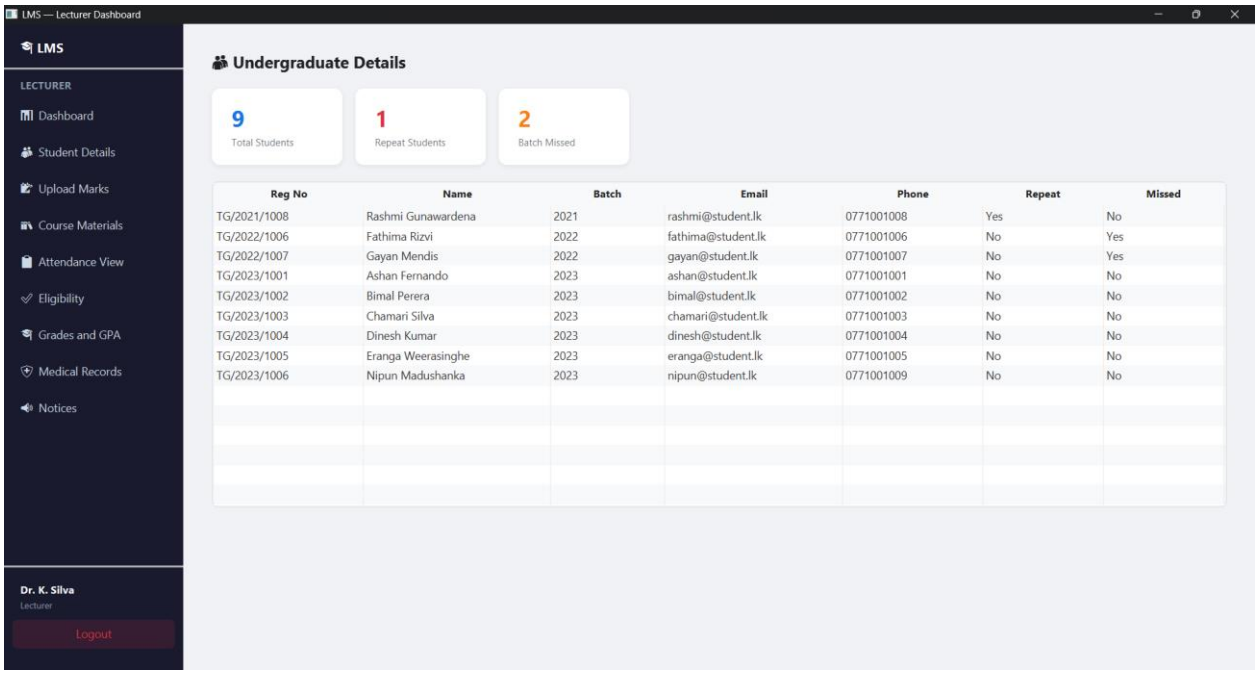
Role
Lecturer
Recent Notices
hi
im perera


LMS — Lecturer Dashboard
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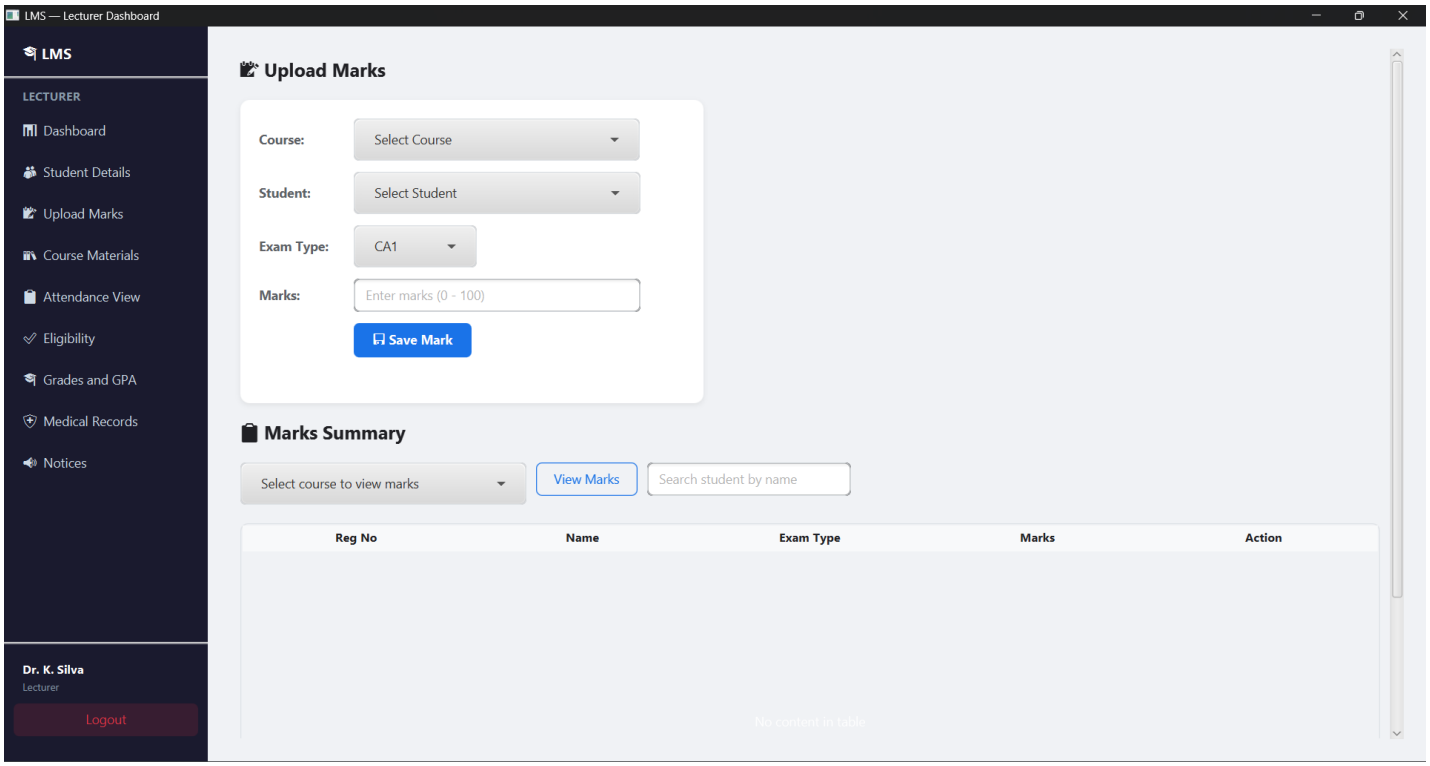
LECTURER
Dashboard
Student Details
Upload Marks
Course Materials
Attendance View
Eligibility
Grades and GPA
Medical Records
Notices
Dr. K. Silva
Lecturer
Logout
Undergraduate Details
9
Total Students
1
Repeat Students
2
Batch Missed
Reg No | Name | Batch | Email | Phone | Repeat | Missed
TG/2021/1008 | Rashmi Gunawardena | 2021 | rashmi@student.lk | 0771001008 | Yes | No
TG/2022/1006 | Fathima Rizvi | 2022 | fathima@student.lk | 0771001006 | No | Yes
TG/2022/1007 | Gayan Mendis | 2022 | gayan@student.lk | 0771001007 | No | Yes
TG/2023/1001 | Ashan Fernando | 2023 | ashan@student.lk | 0771001001 | No | No
TG/2023/1002 | Bimal Perera | 2023 | bimal@student.lk | 0771001002 | No | No
TG/2023/1003 | Chamari Silva | 2023 | chamari@student.lk | 0771001003 | No | No
TG/2023/1004 | Dinesh Kumar | 2023 | dinesh@student.lk | 0771001004 | No | No
TG/2023/1005 | Eranga Weerasinghe | 2023 | eranga@student.lk | 0771001005 | No | No
TG/2023/1006 | Nipun Madushanka | 2023 | nipun@student.lk | 0771001009 | No | No


LMS — Lecturer Dashboard
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Dashboard
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Course Materials
Attendance View
Eligibility
Grades and GPA
Medical Records
Notices
Dr. K. Silva
Lecturer
Logout
Upload Marks
Course:
Select Course
Student:
Select Student
Exam Type:
CA1
Marks:
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Save Mark
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Select course to view marks
View Marks
Search student by name
Reg No | Name | Exam Type | Marks | Action
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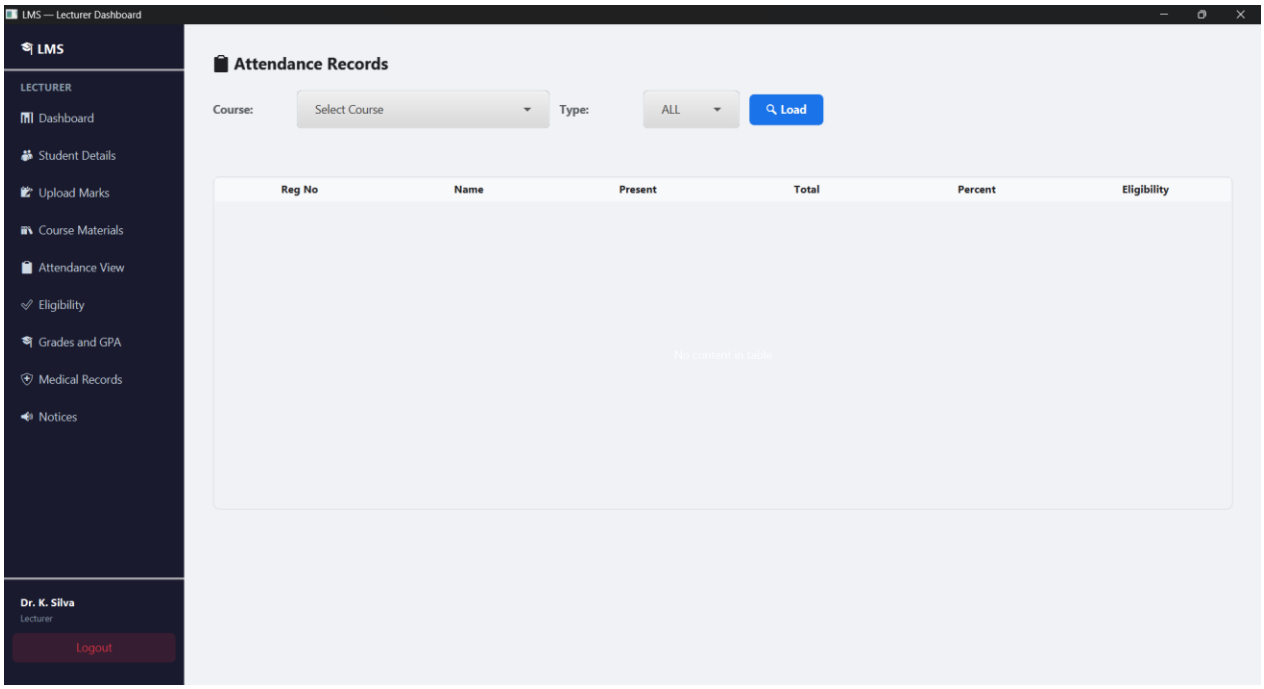
LMS — Lecturer Dashboard
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Dashboard
Student Details
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Course Materials
Attendance View
Eligibility
Grades and GPA
Medical Records
Notices
Dr. K. Silva
Lecturer
Logout
Attendance Records
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Type:
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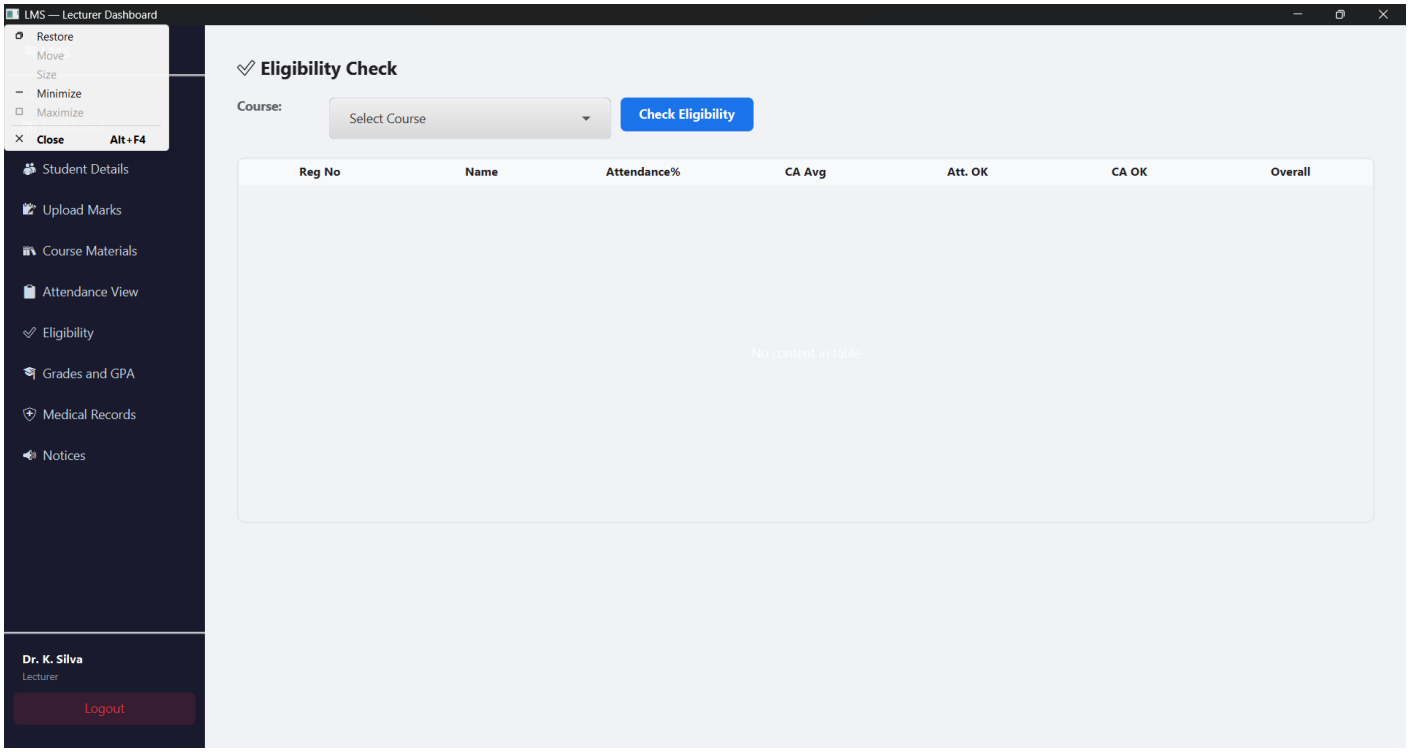
Load
Reg No
Name
Present
Total
Percent
Eligibility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Minimize
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Dr. K. Silva
Lecturer
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Course:
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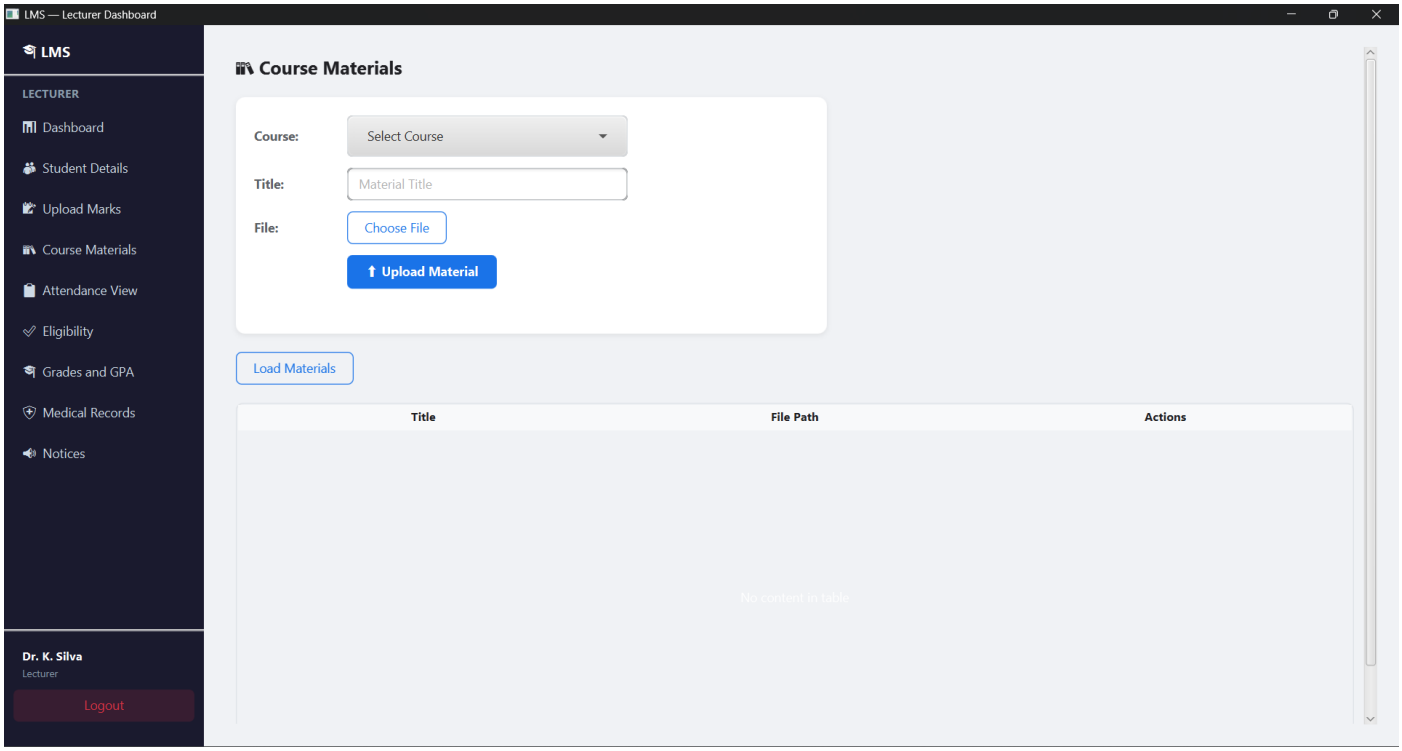
Reg No
Name
Attendance%
CA Avg
Att. OK
CA OK
Overall


LMS — Lecturer Dashboard
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Course Materials
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Notices
Dr. K. Silva
Lecturer
Logout
Course Materials
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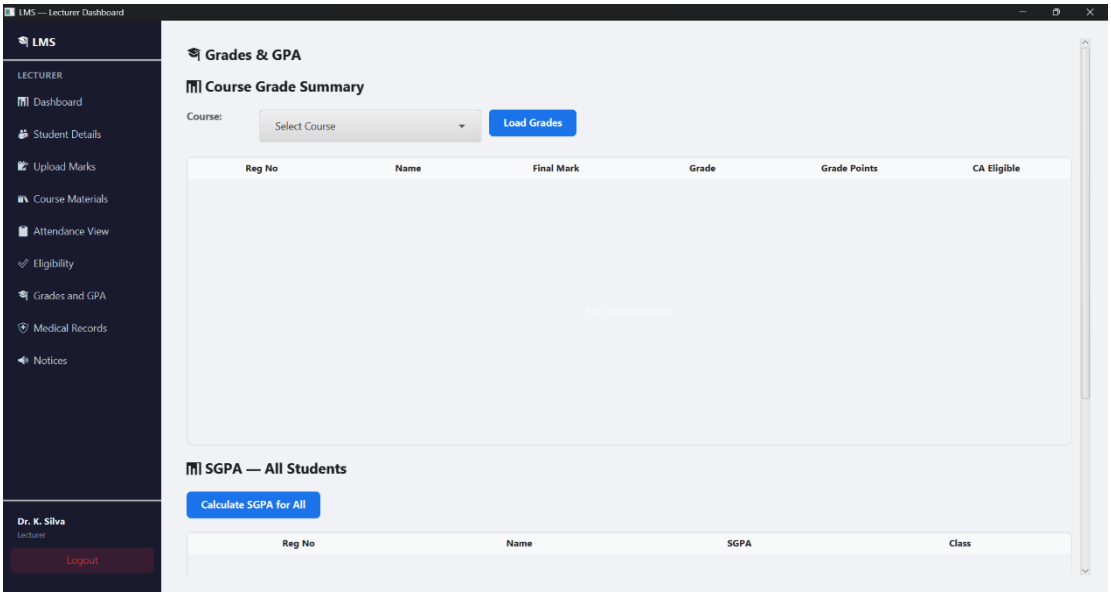
Material Title
File:
Choose File
Upload Material
Load Materials
Title
File Path
Actions


LMS — Lecturer Dashboard
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Dashboard
Student Details
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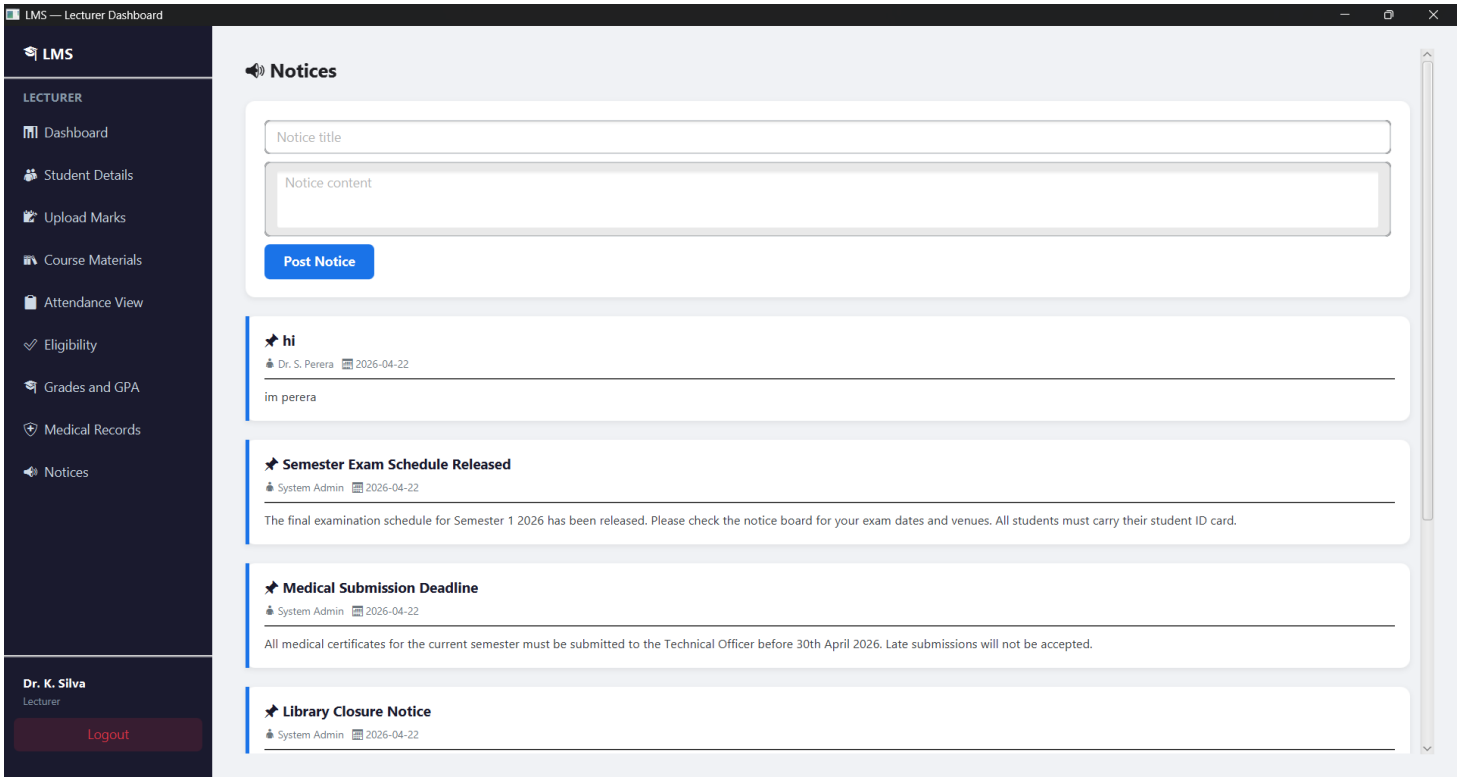
Course Materials
Attendance View
Eligibility
Grades and GPA
Medical Records
Notices
Dr. K. Silva
Lecturer
Logout
Grades & GPA
Course Grade Summary
Course:
Select Course
Load Grades
Reg No
Name
Final Mark
Grade
Grade Points
CA Eligible
SGPA — All Students
Calculate SGPA for All
Reg No
Name
SGPA
Class


LMS — Lecturer Dashboard
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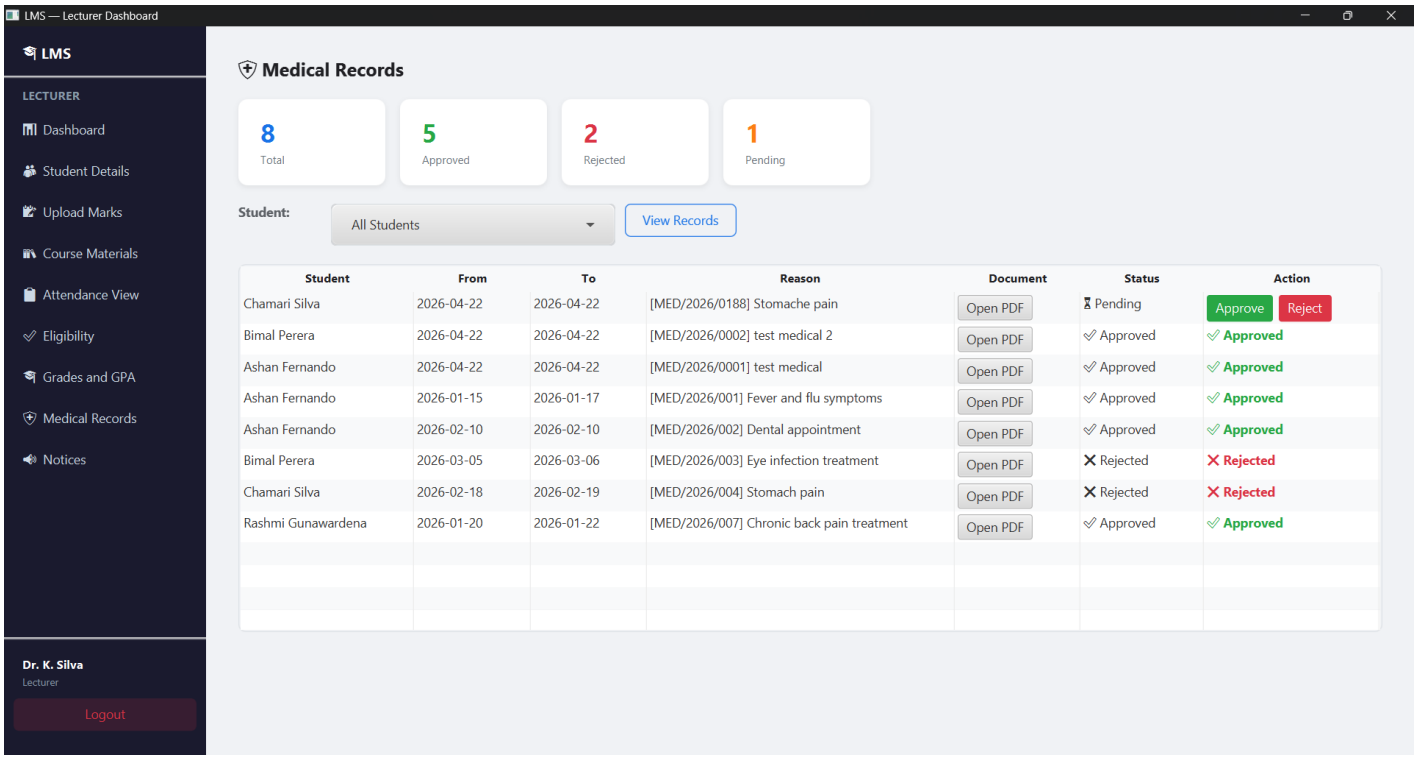
Student Details
Upload Marks
Course Materials
Attendance View
Eligibility
Grades and GPA
Medical Records
Notices
Dr. K. Silva
Lecturer
Logout
Notices
Notice title
Notice content
Post Notice
hi
Dr. S. Perera 2026-04-22
im perera
Semester Exam Schedule Released
System Admin 2026-04-22
The final examination schedule for Semester 1 2026 has been released. Please check the notice board for your exam dates and venues. All students must carry their student ID card.
Medical Submission Deadline
System Admin 2026-04-22
All medical certificates for the current semester must be submitted to the Technical Officer before 30th April 2026. Late submissions will not be accepted.
Library Closure Notice
System Admin 2026-04-22


LMS — Lecturer Dashboard
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Student Details
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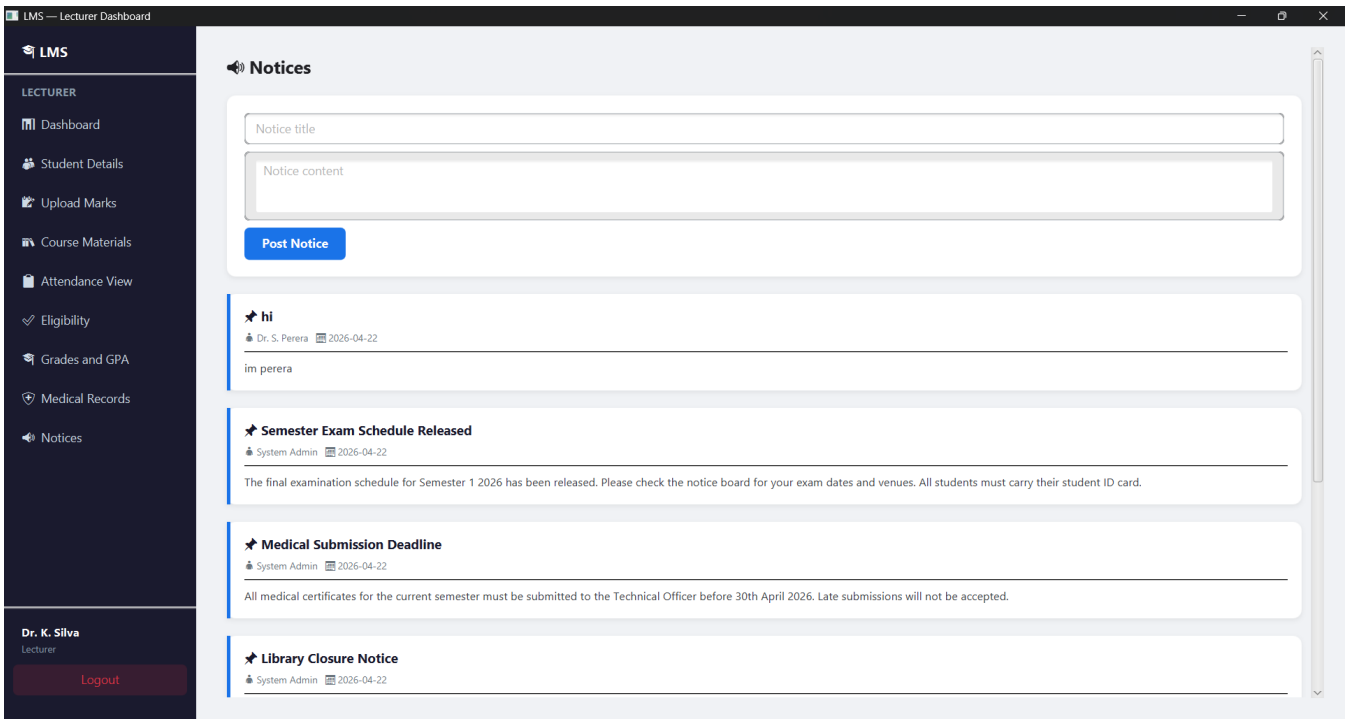
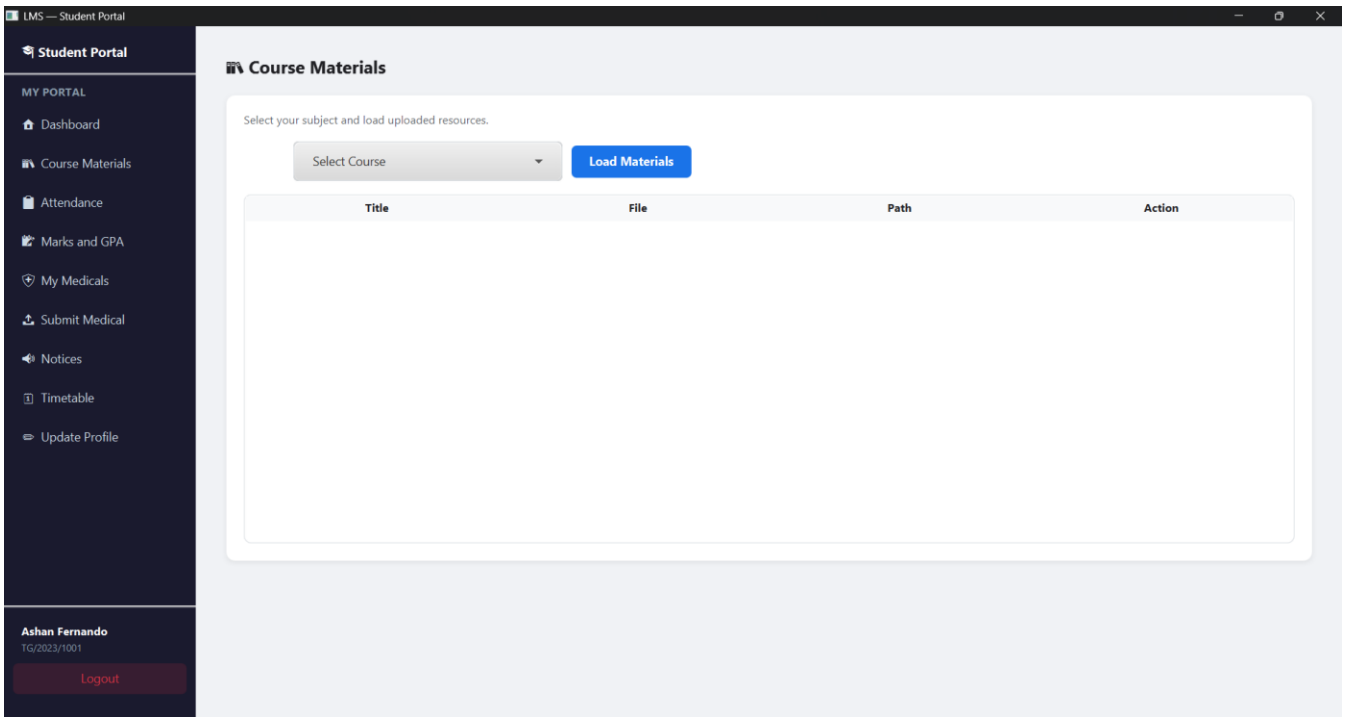
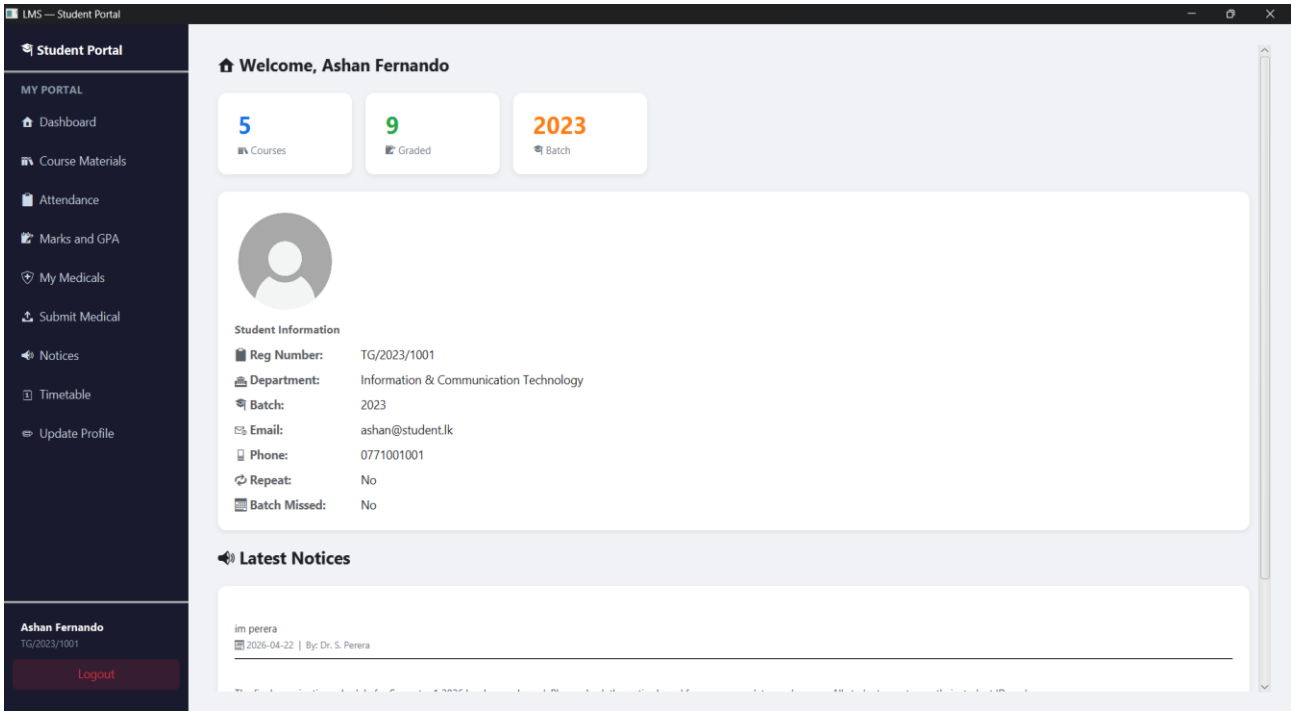
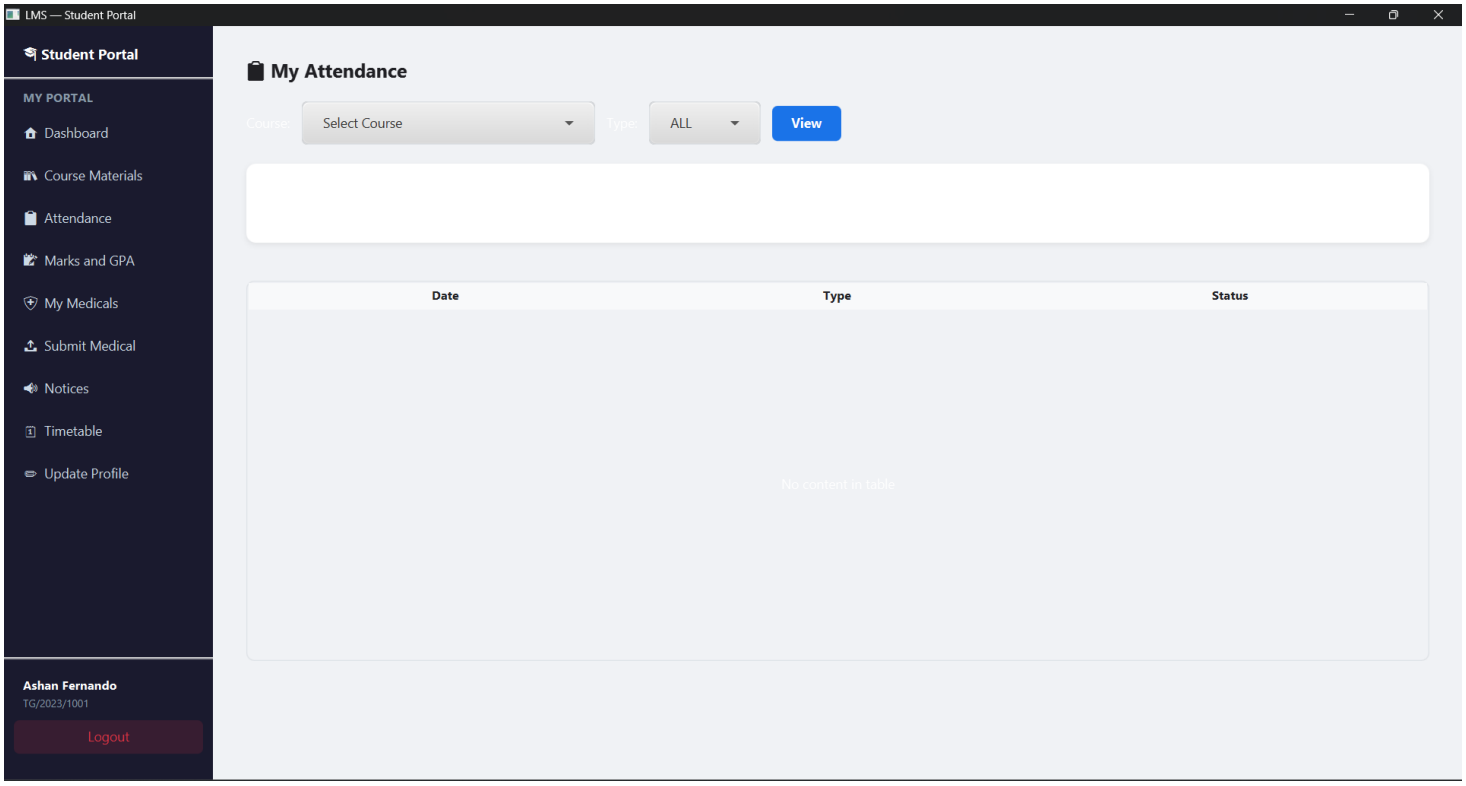
Course Materials
Attendance View
Eligibility
Grades and GPA
Medical Records
Notices
Dr. K. Silva
Lecturer
Logout
Medical Records
8 Total
5 Approved
2 Rejected
1 Pending
Student:
All Students
View Records
Student | From | To | Reason | Document | Status | Action
Chamari Silva | 2026-04-22 | 2026-04-22 | [MED/2026/0188] Stomache pain | Open PDF | Pending | Approve Reject
Bimal Perera | 2026-04-22 | 2026-04-22 | [MED/2026/0002] test medical 2 | Open PDF | Approved | Approved
Ashan Fernando | 2026-04-22 | 2026-04-22 | [MED/2026/0001] test medical | Open PDF | Approved | Approved
Ashan Fernando | 2026-01-15 | 2026-01-17 | [MED/2026/001] Fever and flu symptoms | Open PDF | Approved | Approved
Ashan Fernando | 2026-02-10 | 2026-02-10 | [MED/2026/002] Dental appointment | Open PDF | Approved | Approved
Bimal Perera | 2026-03-05 | 2026-03-06 | [MED/2026/003] Eye infection treatment | Open PDF | Rejected | Rejected
Chamari Silva | 2026-02-18 | 2026-02-19 | [MED/2026/004] Stomach pain | Open PDF | Rejected | Rejected
Rashmi Gunawardena | 2026-01-20 | 2026-01-22 | [MED/2026/007] Chronic back pain treatment | Open PDF | Approved | Approved



Undergraduate




LMS — Student Portal
Student Portal
MY PORTAL
Dashboard
Course Materials
Attendance
Marks and GPA
My Medicals
Submit Medical
Notices
Timetable
Update Profile
Ashan Fernando
TG/2023/1001
Logout
Welcome, Ashan Fernando
5
Courses
9
Graded
2023
Batch
Student Information
Reg Number: TG/2023/1001
Department: Information & Communication Technology
Batch: 2023
Email: ashan@student.lk
Phone: 0771001001
Repeat: No
Batch Missed: No
Latest Notices
im perera
2026-04-22 | By: Dr. S. Perera


LMS — Student Portal
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MY PORTAL
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My Medicals
Submit Medical
Notices
Timetable
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Ashan Fernando
TG/2023/1001
Logout
My Attendance
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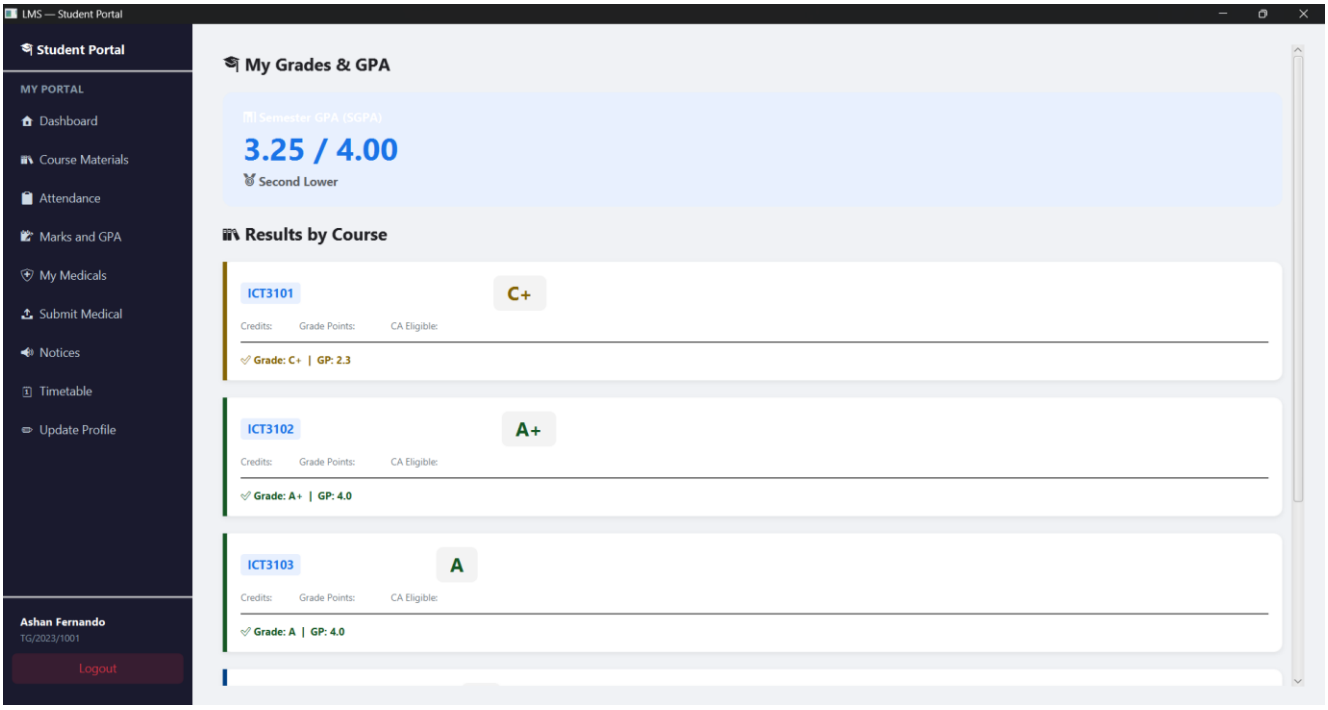
ALL
View
Date
Type
Status


LMS — Student Portal
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Course Materials
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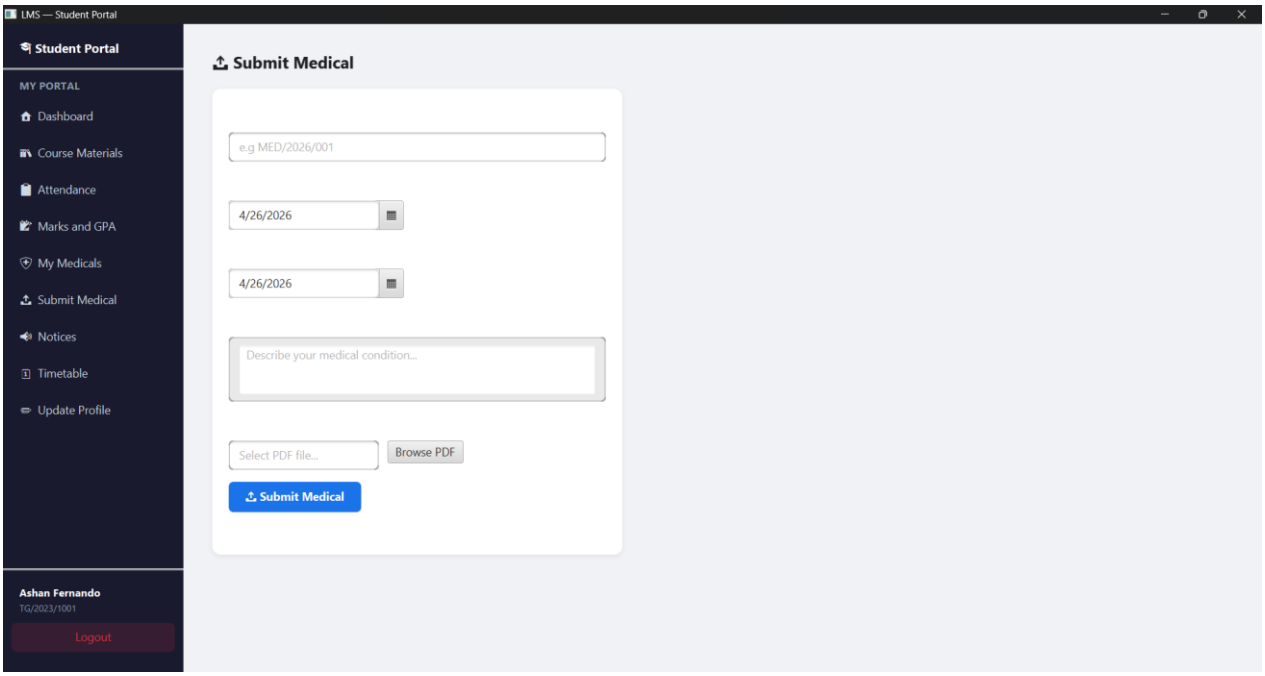
Marks and GPA
My Medicals
Submit Medical
Notices
Timetable
Update Profile
Ashan Fernando
TG/2023/1001
Logout
My Grades & GPA
Semester GPA (SGPA)
3.25 / 4.00
Second Lower
Results by Course
ICT3101
C+
Credits:
Grade Points:
CA Eligible:
Grade: C+ | GP: 2.3
ICT3102
A+
Credits:
Grade Points:
CA Eligible:
Grade: A+ | GP: 4.0
ICT3103
A
Credits:
Grade Points:
CA Eligible:
Grade: A | GP: 4.0


LMS — Student Portal
Student Portal
MY PORTAL
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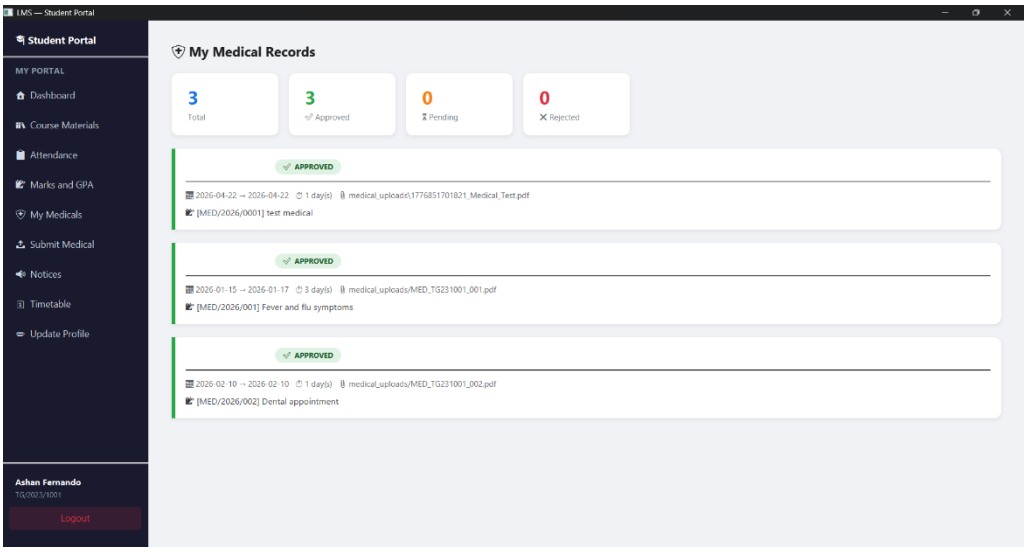
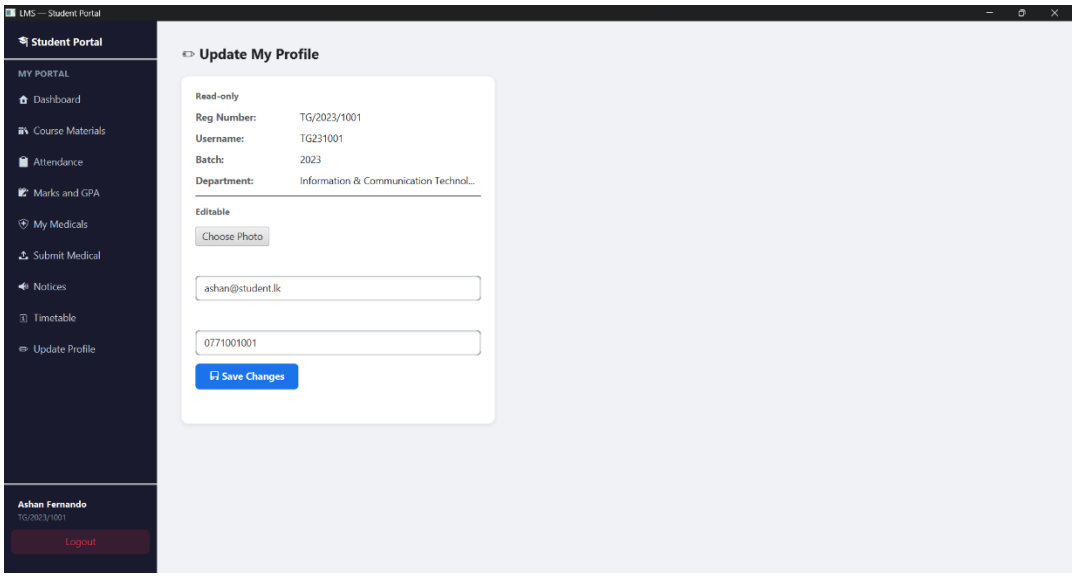
Course Materials
Attendance
Marks and GPA
My Medicals
Submit Medical
Notices
Timetable
Update Profile
Ashan Fernando
TG/2023/1001
Logout
Submit Medical
e.g MED/2026/001
4/26/2026
4/26/2026
Describe your medical condition...
Select PDF file...
Browse PDF
Submit Medical





Tools and Techniques we have used for this project

- IntelliJ idea and JavaFX UI
  - To create and maintain GUI (Graphical User Interface)
  - To maintain java codes
  - To Connect the database using MySQL connectors

- MYSQL server
  - To create and manage database efficiency
  - To create table structures and data insertion
- Draw.io
  - Used to create ER diagram and Class diagram
- GitHub and GitHub Desktop
  - To manage and collaborate on database and version control
- Microsoft word
  - To Create Final Report and SRS report

Security measures that we have taken to project

- **Role-based access control (RBAC)** to restrict access based on user roles.
- **Database constraints** (Primary key, foreign keys, Auto increment) to maintain data integrity.
- **Limited profile editing permissions** (e.g.,. Lecturer,Technical officer, students can't change username or password).

Functional Requirements

Admin

- Create and maintain all User Profiles
- Create and maintain Courses
- Create and maintain Notices

Lecturer

- Can update their profile except username and password
- Can modify and add materials to courses
- Can upload marks for all kind of exams
- Can see undergraduate details
- Can see undergraduate eligibility
- Can see undergraduate marks, grades and GPA
- Can see attendance and medical records of undergraduate
- Can see notices
- Create and maintain Timetables

Technical Officer

- Can update their profile except username and password
- Can add and maintain attendance details of undergraduate
- Can add and maintain medical details of undergraduate
- Can see notices
- Can see timetables of their department

Undergraduate

- Can update only contact details and profile picture of their profile
- Can see attendance details
- Can see medical details
- Can see their course details
- Can see their grades and GPA
- Can see their timetables
- Can see notices

Non Functional Requirements

1. **Performance Requirements**

- The system must support simultaneous access for more users without performance delay.

2. **Security Requirements**

- Only authorized users can access specific functionalities based on their roles (Admin, Lecturer, Technical Officer, Undergraduate).
- Sensitive data like passwords must not be displayed or logged in any format.

3. **Maintainability**

- The code must be organized following object-oriented programming (OOP) principles (such as class, object encapsulation, polymorphism , Inheritance) for easy maintenance.

4. **Portability**

- The system should be able to run on Windows, Linux, and macOS operating systems .

Problems that we faced during the development of the Solution.

- **Database complexity**: Mapping relationships among multiple entities (e.g., medical,attendance) was challenging.
- **GUI synchronization**: Ensuring that the user interface reflects real-time updates from the database
- **Team coordination**: Dividing tasks and merging contributions without conflicts was time-consuming.

Solutions / How we have overcome the above the identified problems

- **Database complexity**: We created a detailed ER diagram to represent TechLMS details.
- **GUI Synchronization**: We used a single main frame with multiple panels, switching panels inside the main frame based on user actions to keep the interface simple and organized.
- **Team coordination**: Used GitHub for version control, assigned tasks to project , and conducted regular group communication.

Individual contribution to System development

TG_no	Name	Individual contribution
TG/2023/1715	A.P.G.H.Pramoda	*Create SRS Document *Create Class diagram *Grade Calculator *Notice Functionalities *Mark Functionalities *Medical Functionalities
TG/2023/1758	A.M.I.T.B.Adhikari	* Admin Functionalities *User login page *Lecturer Functionalities *Create ER Diagram *Create Final Report
TG/2023/1732	J.L.A.L.Sewwandi	*Undergraduate Functionalities *Technical Officer Functionalities *Attendance Functionalities *Password Functionalities
TG/2023/1742	N.V.T.J.Gunasekara	*Course Functionalities *Time table Functionalities *Database Connection and SQL file

GitHub Repository Link: https://github.com/IndusaraAdhikari003/Java-Mini-Project.git

References

- W3 Schools
- Lecture Notes
- Oracle Java Documentation